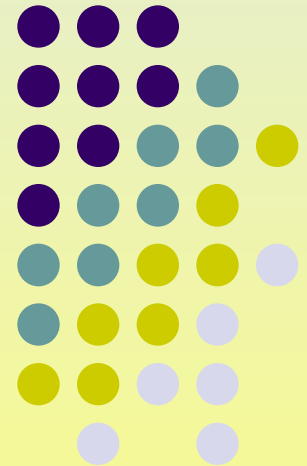


Romania Low Carbon and Green Growth Study

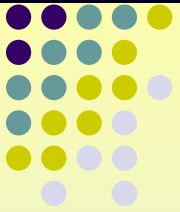
Energy Sector: End-use Energy Demand Projection

**Sunil Malla, Govinda R. Timilsina and
Oluwafemi Faleye**

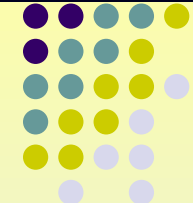
Bucharest, Romania
16 February, 2015



Presentation Outline



- 1. Overview of current energy demand situation in Romania**
- 2. End-use energy demand projection**
 - 2.1 Methodology**
 - 2.2 Data and assumptions**
 - 2.3 Key results**
- 3. Concluding remarks**



1. Overview of current energy demand situation in Romania

2. End-use energy demand projection

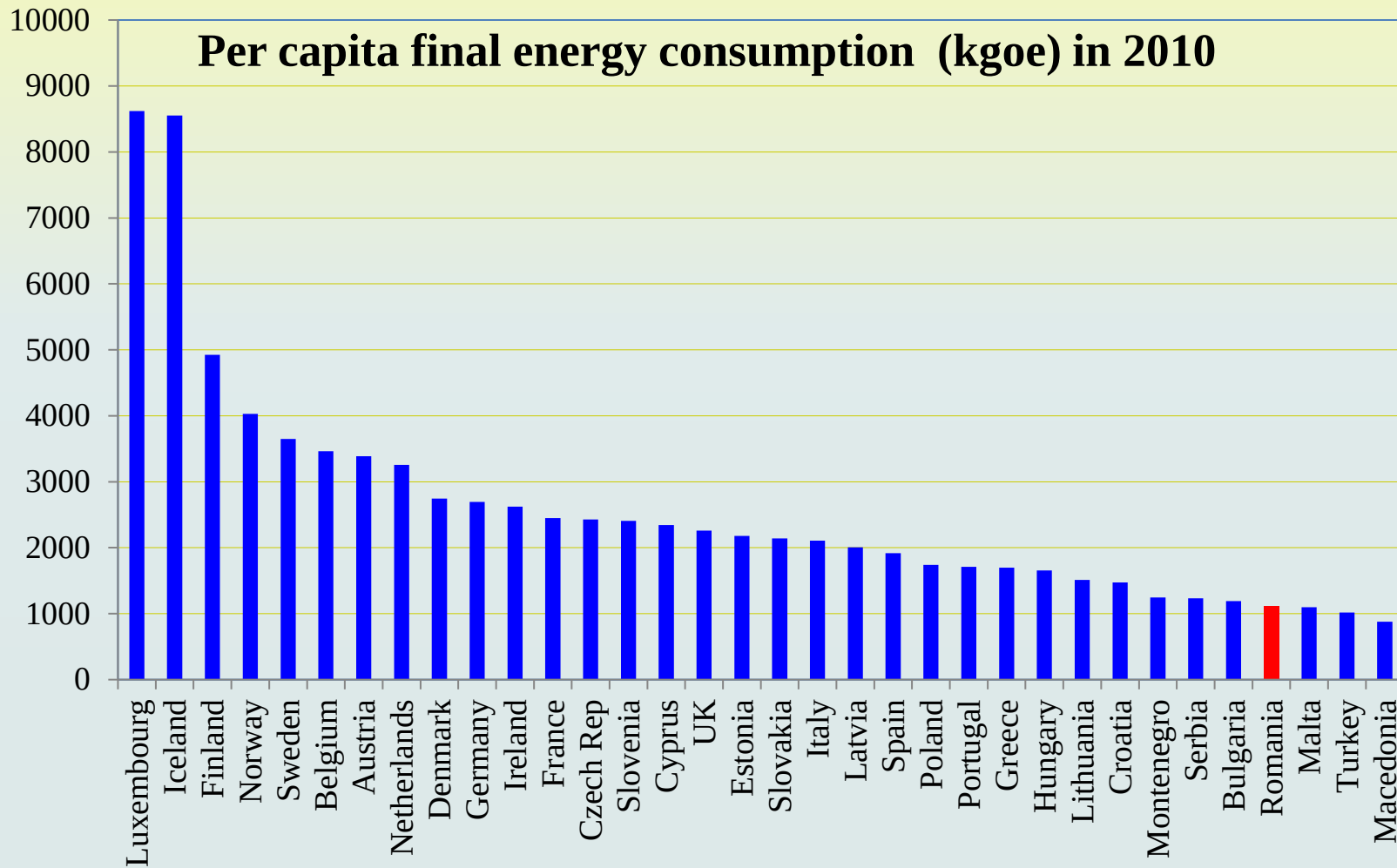
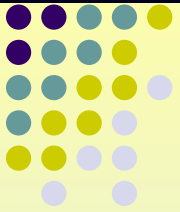
2.1 Methodology

2.2 Data and assumptions

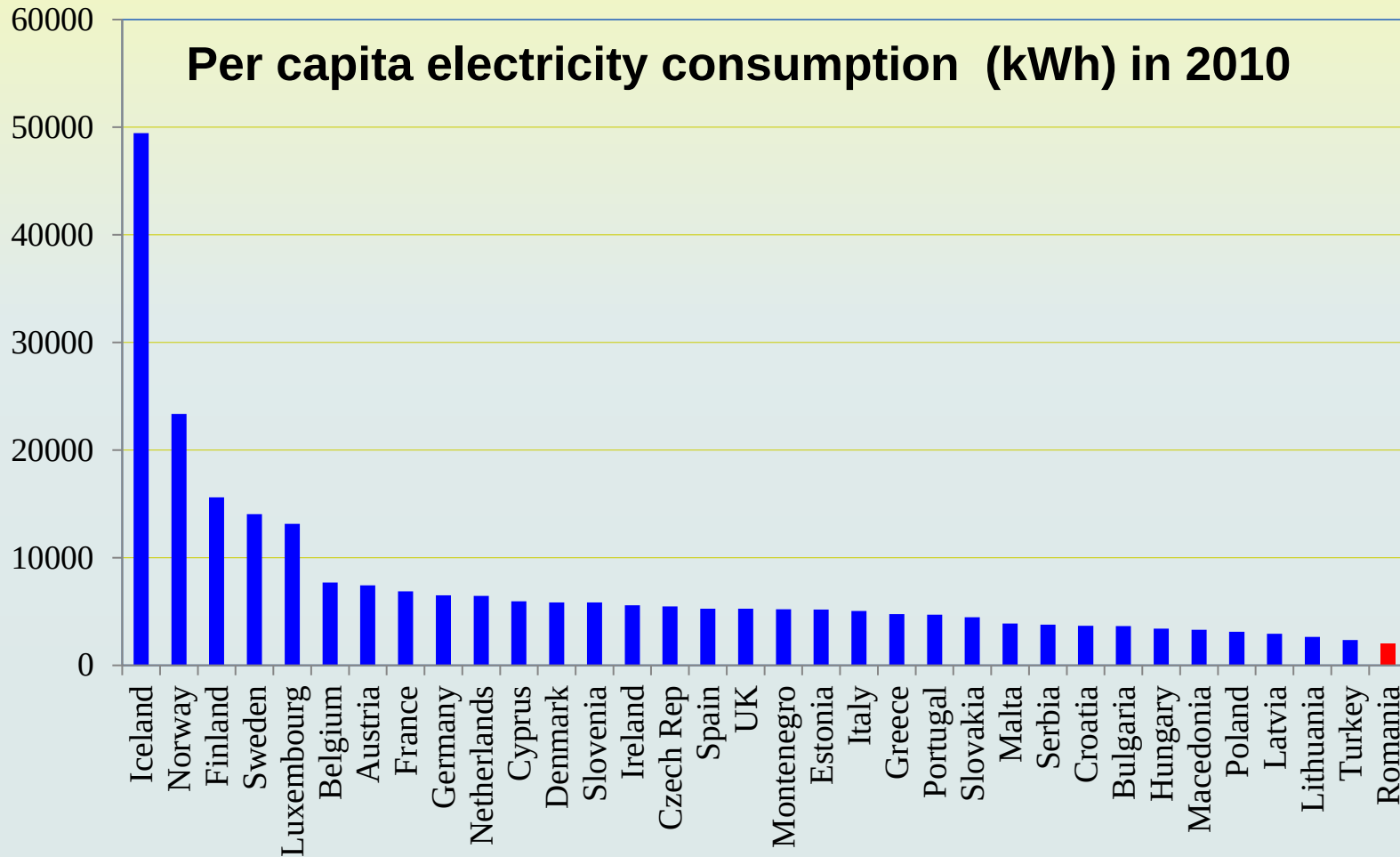
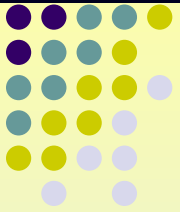
2.3 Key results

3. Concluding remarks

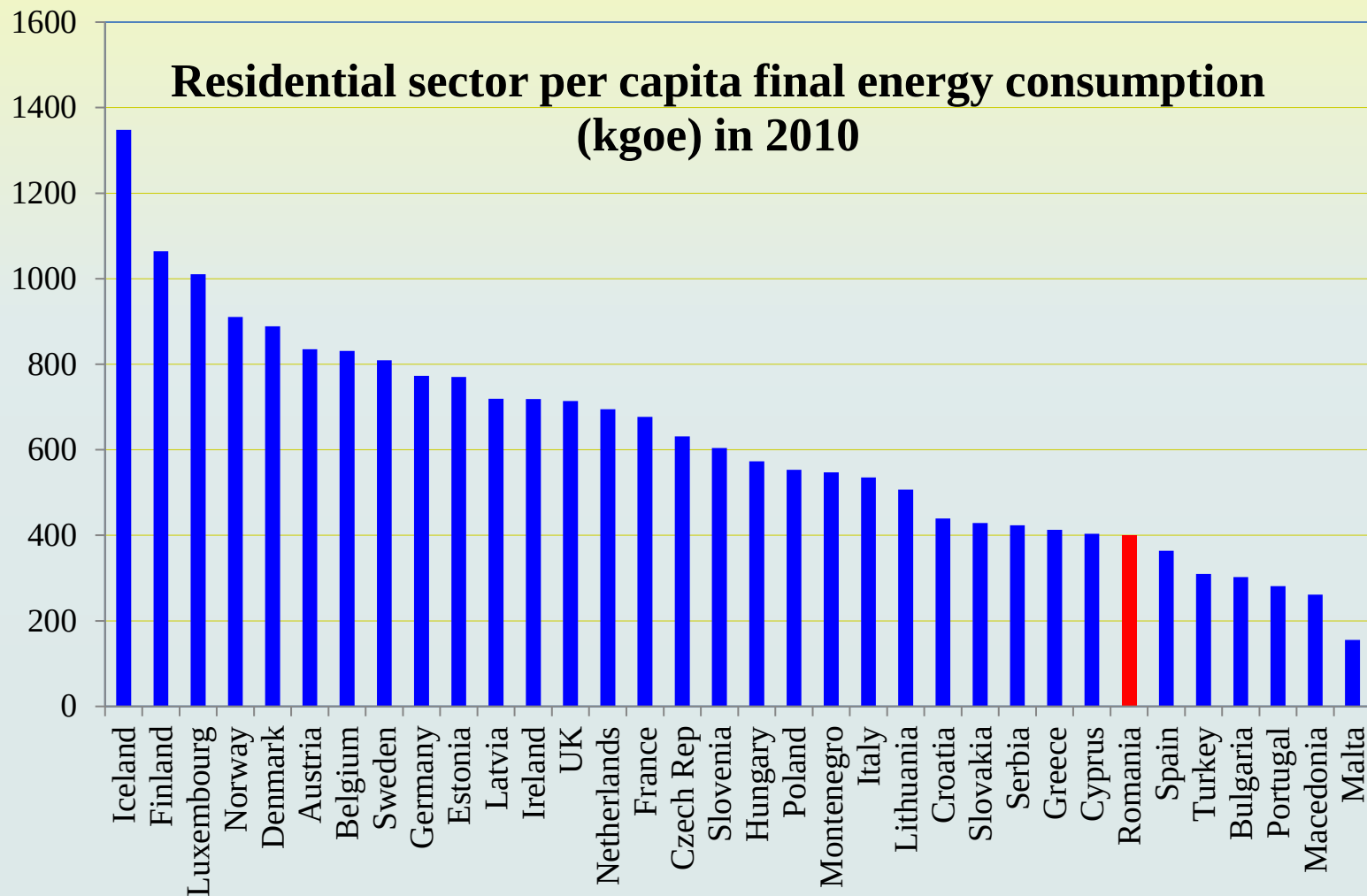
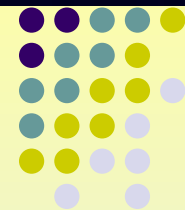
Energy Indicators in Romania and EU in 2010



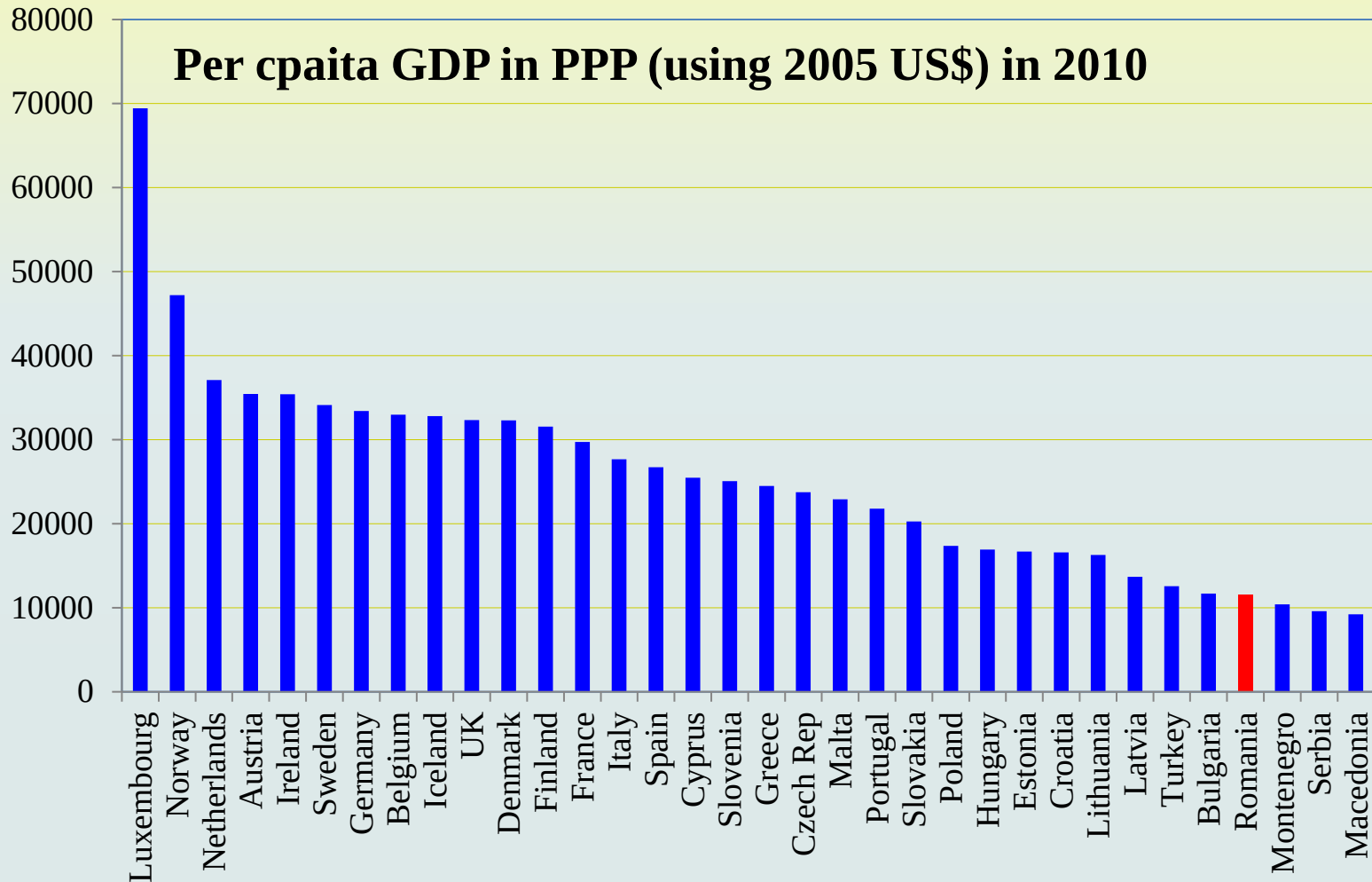
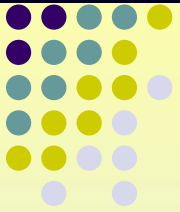
Energy Indicators in Romania and EU in 2010



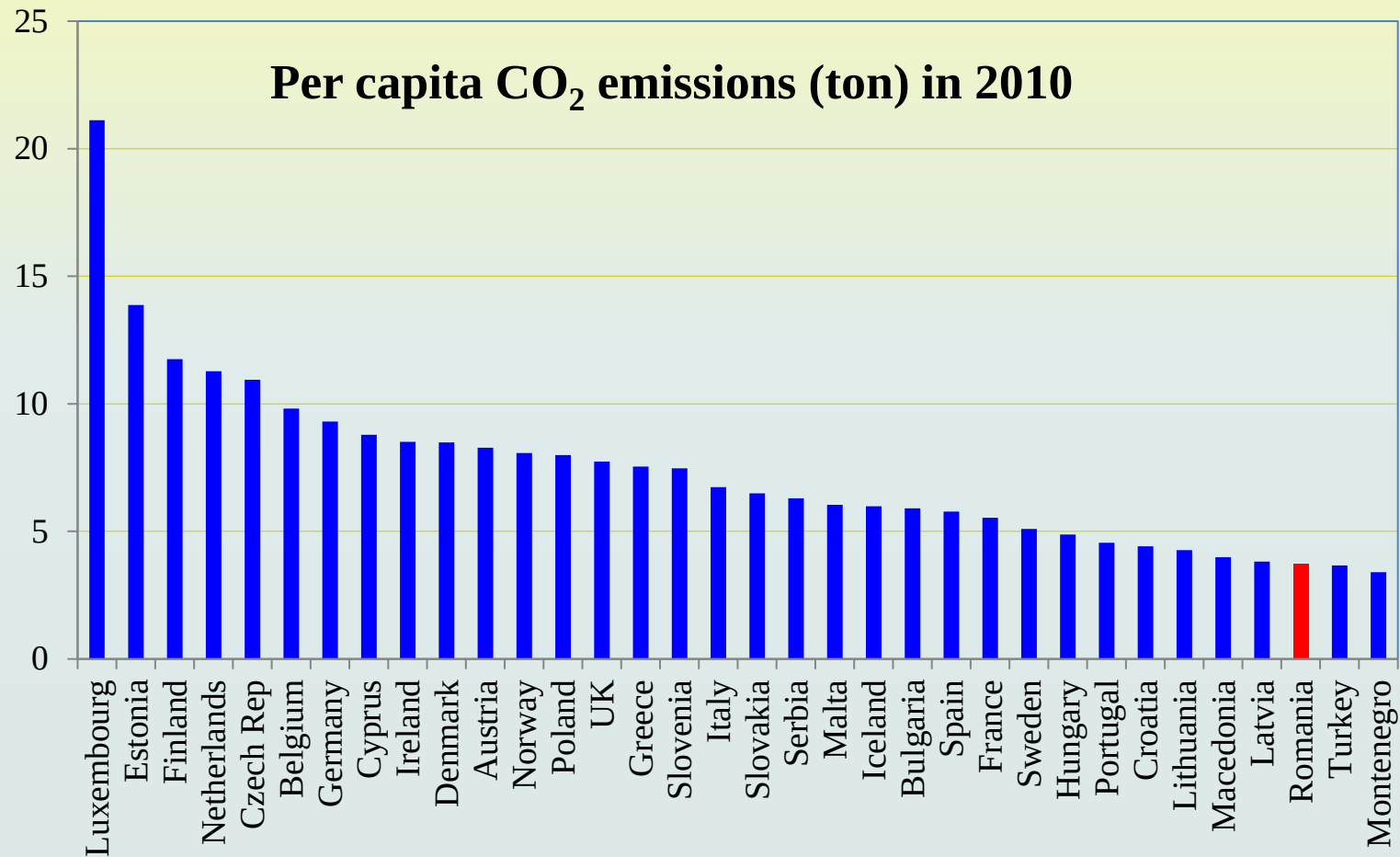
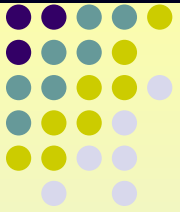
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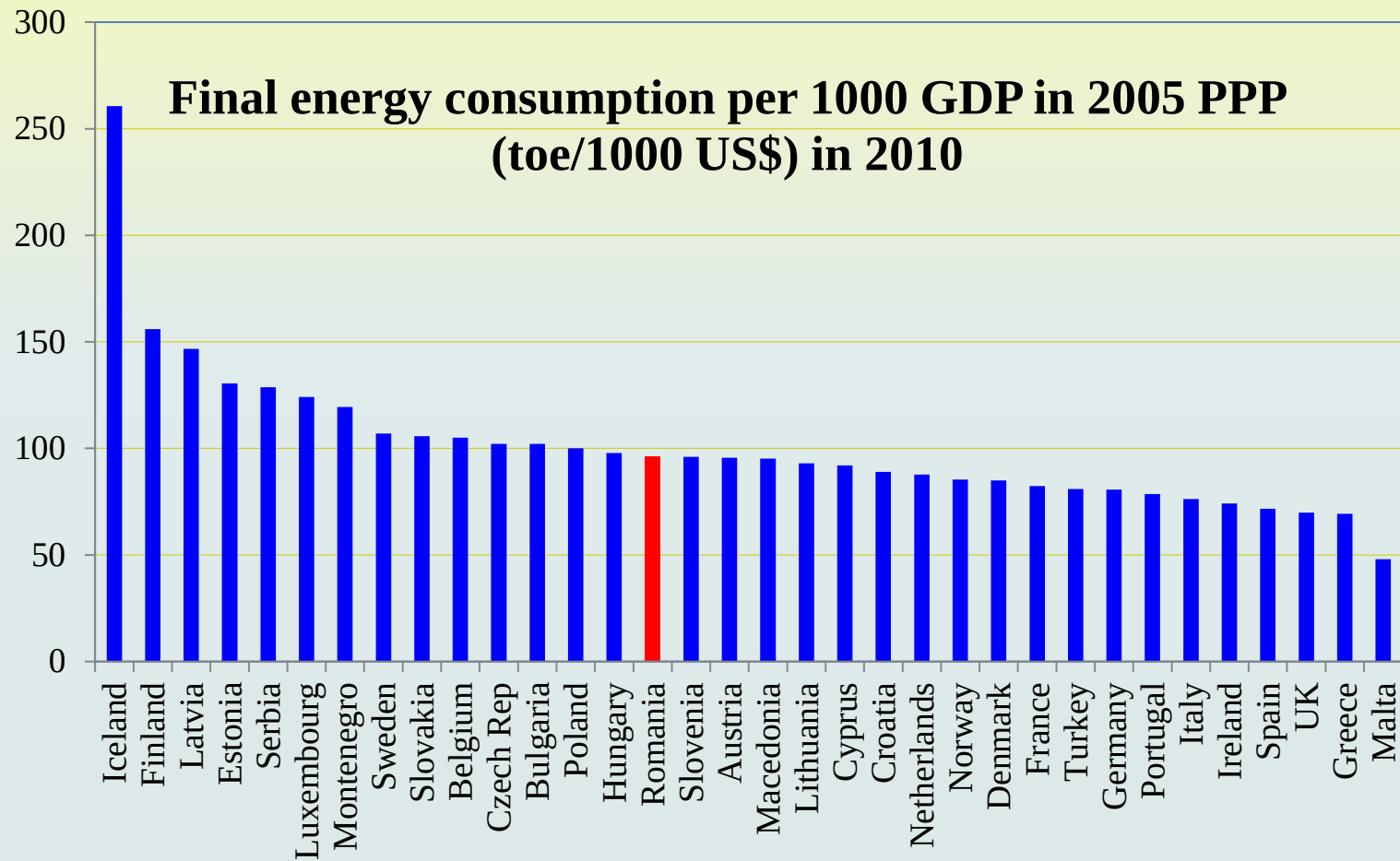
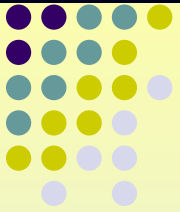
Energy Indicators in Romania and EU in 2010



Energy Indicators in Romania and EU in 2010

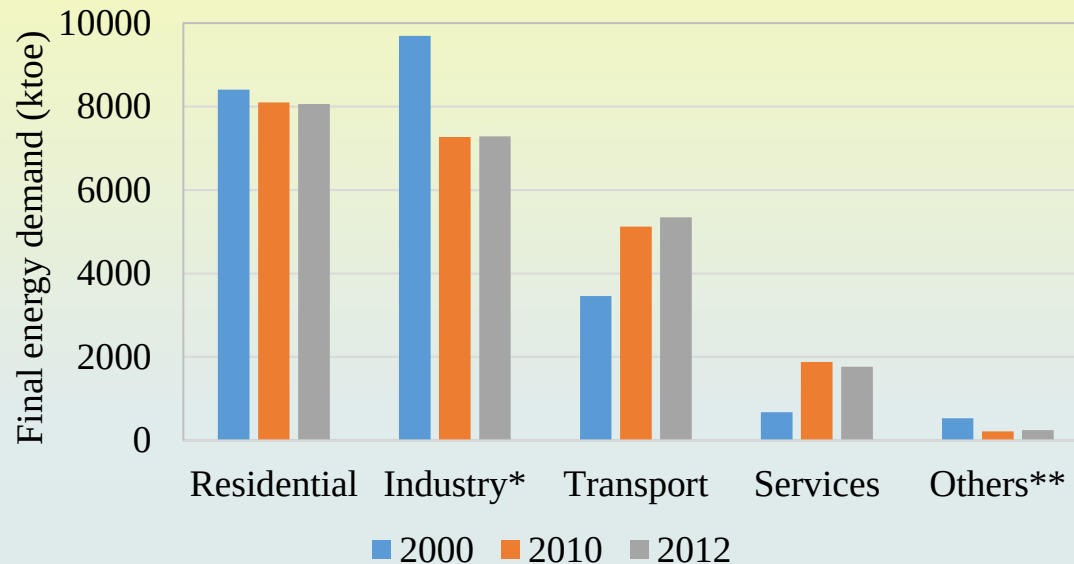


Energy Indicators in Romania and EU in 2010



Historical situation:

Final energy demand (FED) by sector in 2000, 2010 and 2012



Notes:

* Industry also includes agriculture, forestry and fishery.

** Others includes non-specified.

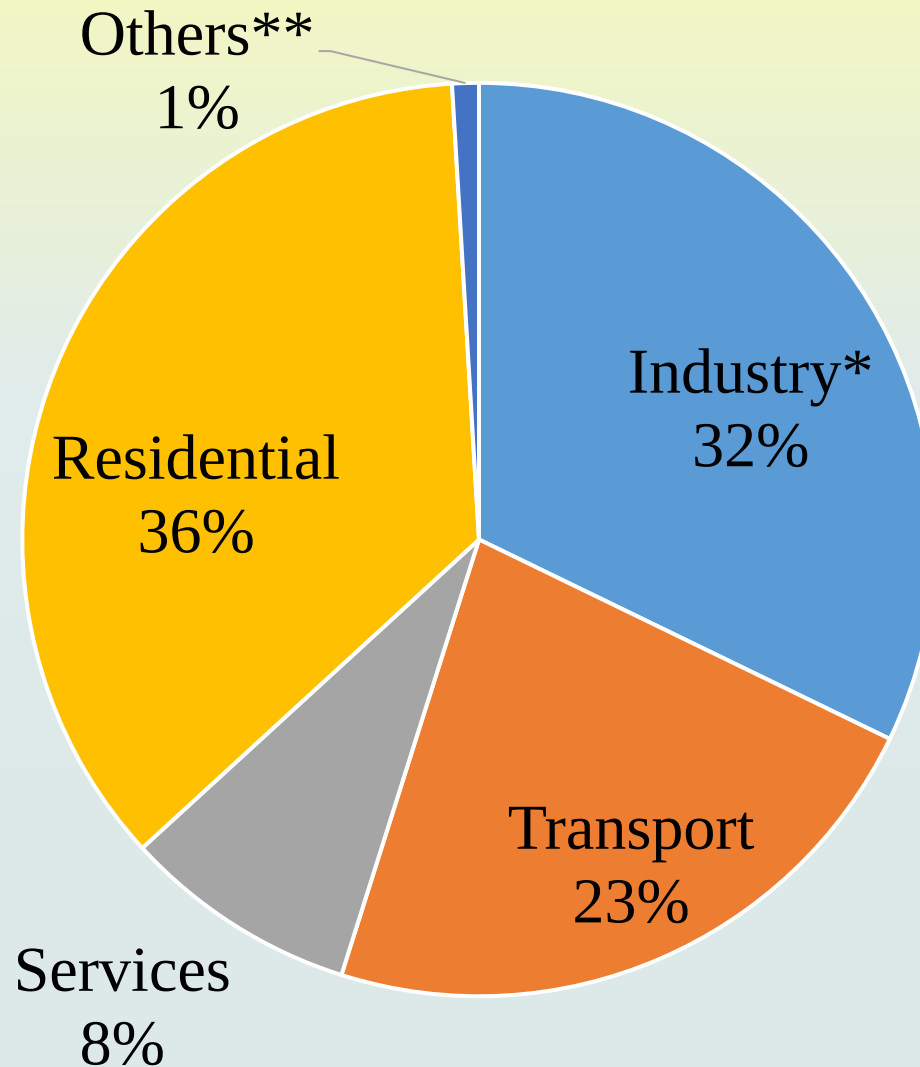
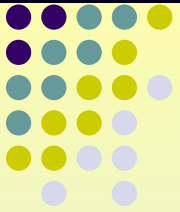
Source: Eurostat

Between 2000 and 2012, total FED is unchanged. However:

- ❑ Industry: - 25% (**decline**)
- ❑ Transport: 54% (**increase**)
- ❑ Services: 162% (**increase**)
- ❑ Residential: - 4% (**decline**)

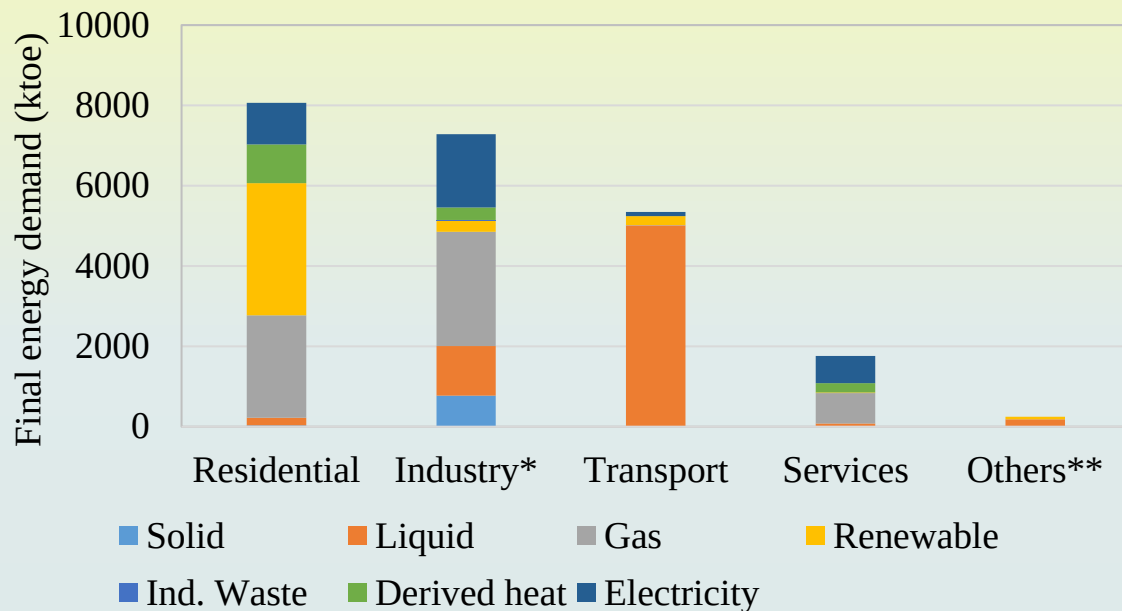
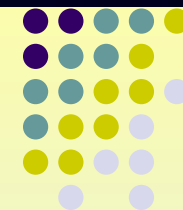
Current situation:

Final energy demand (FED) by sector in 2012



Current situation:

Final energy demand (FED) by sector and by energy type in 2012



Industry:

Gas, electricity and liquid fuels

Transport:

Liquid fuels

Services:

Gas and electricity

Residential:

Renewable (biomass), gas, derived heat and electricity

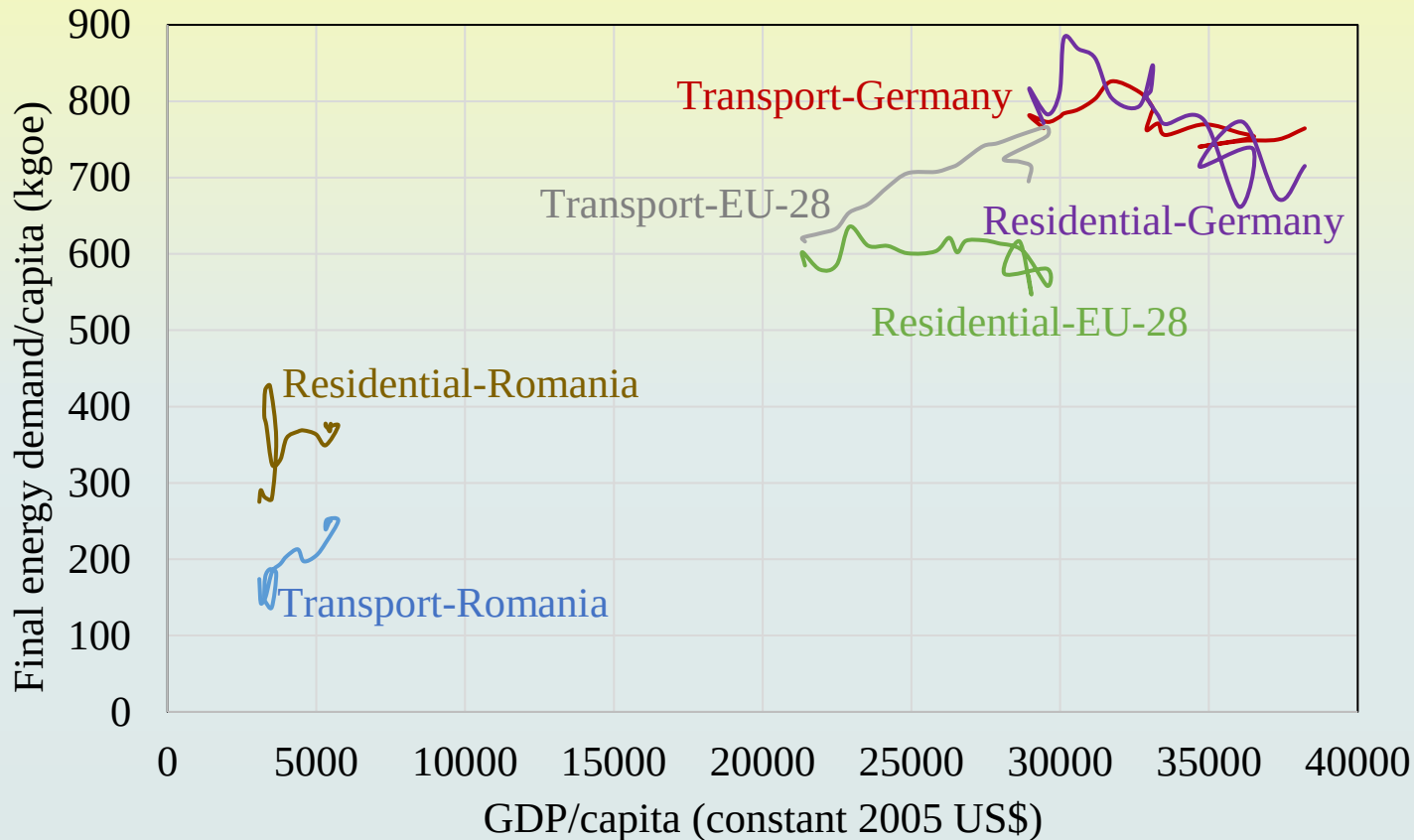
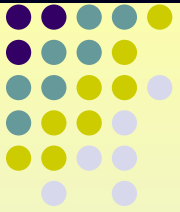
Notes:

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** Others includes non-specified.

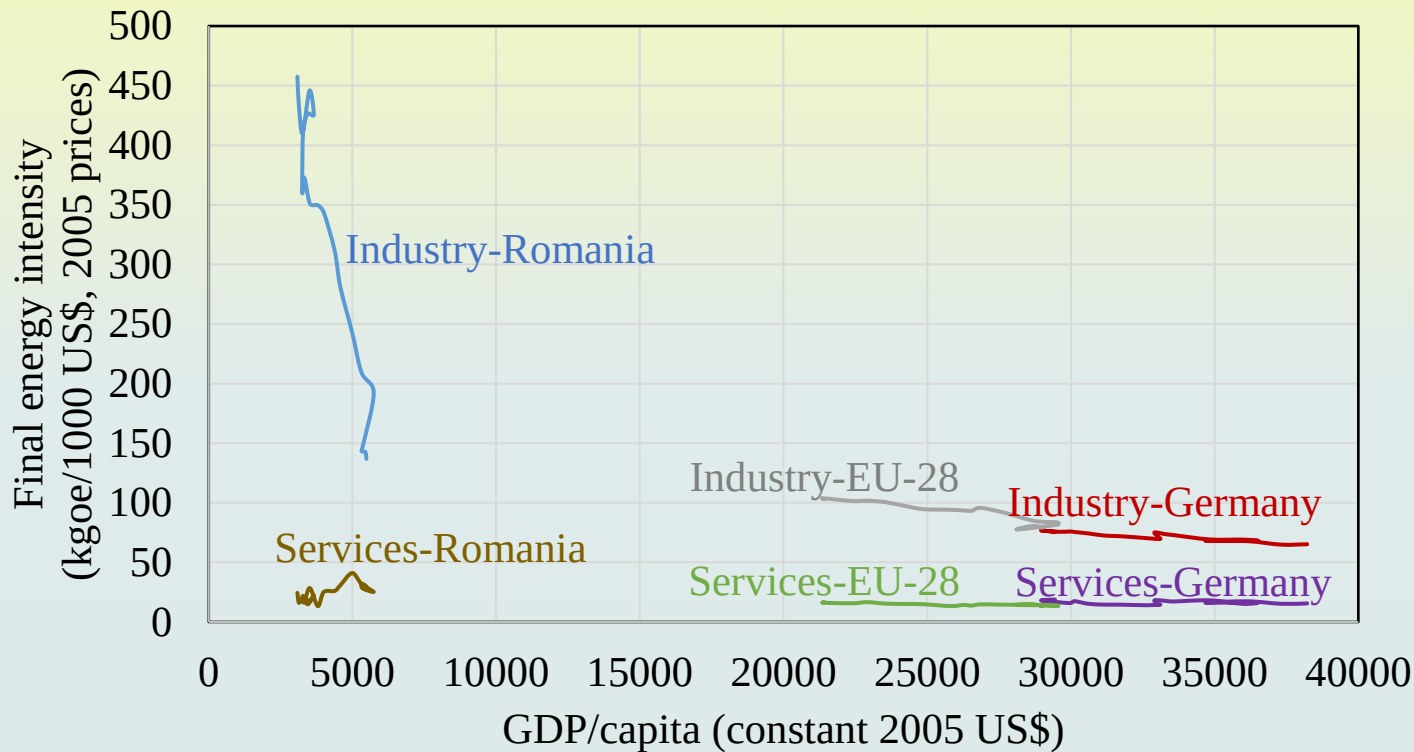
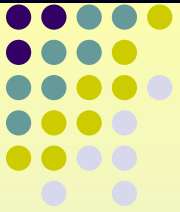
Source: Eurostat

Relation between GDP/capita and final energy demand (FED), 1992-2012

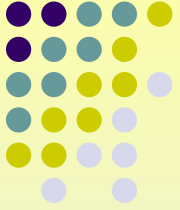


In per capita terms, FED of residential and transport sectors, and income in Romania are much lower compared to Germany and EU-28 average.

Relation between final energy intensity and GDP/capita, 1992-2012



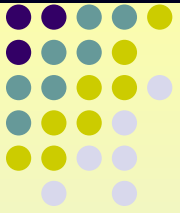
- ❑ Final energy intensity of industry sector relative to GDP per capita is declining in Romania, Germany and the EU-28 average;
- ❑ Final energy intensity is relatively much higher in Romania when compared with Germany and the EU-28 average.



1. Overview of current energy demand situation in Romania
2. **End-use energy demand projection**
 - 2.1 Methodology
 - 2.2 Data and assumptions
 - 2.3 Key results
3. Concluding remarks

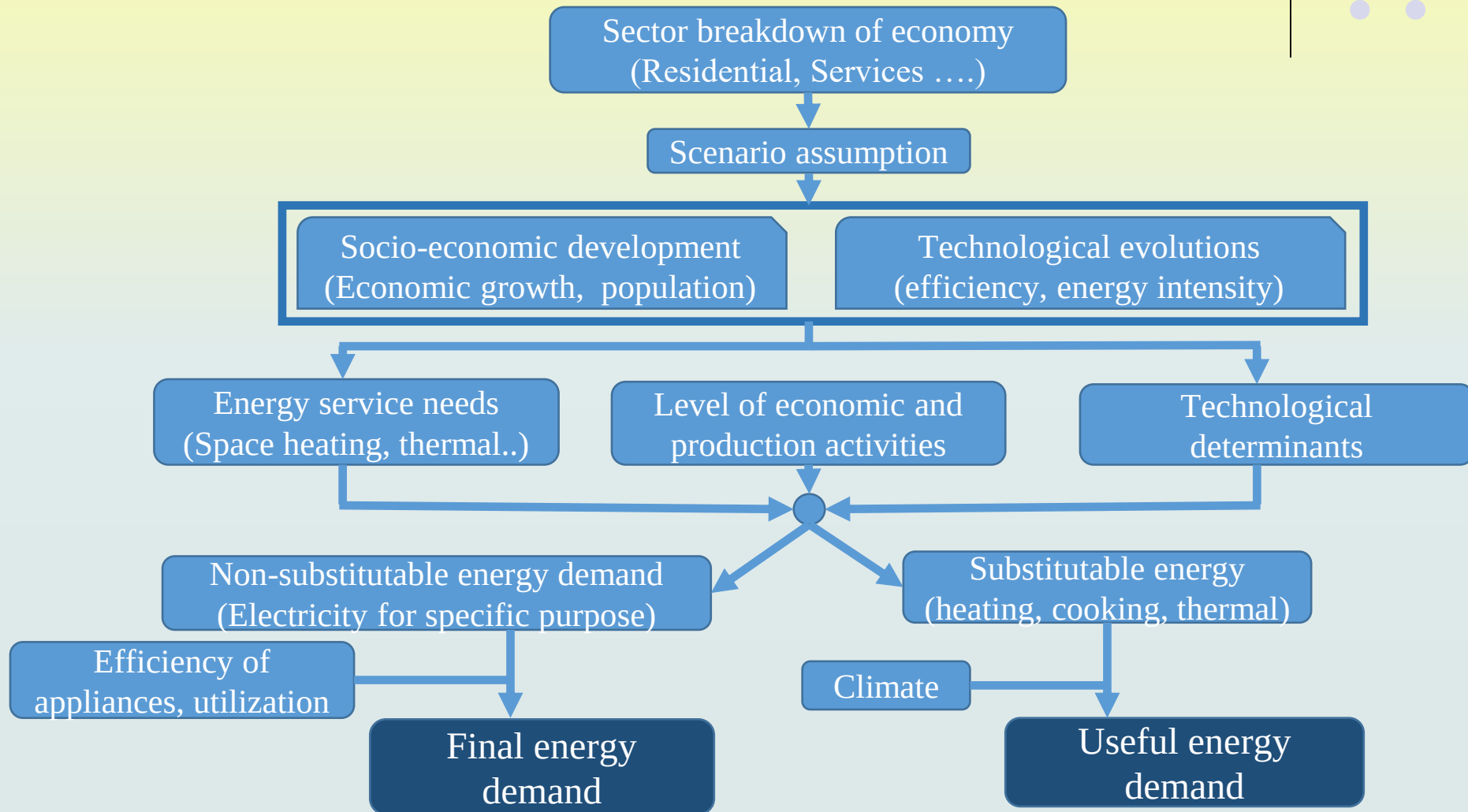
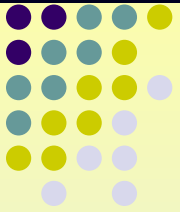
End-use Energy Demand Projection

Methodology – Key Features

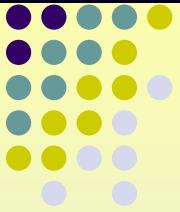


- ☐ Based on bottom-up accounting framework
- ☐ Follows commonly used approach – Activity, Structure and Intensity analysis
- ☐ Built using MS Excel for transparency, flexibility and easy to use
- ☐ Reference year: 2010
- ☐ Time horizon: 40 years up to 2050

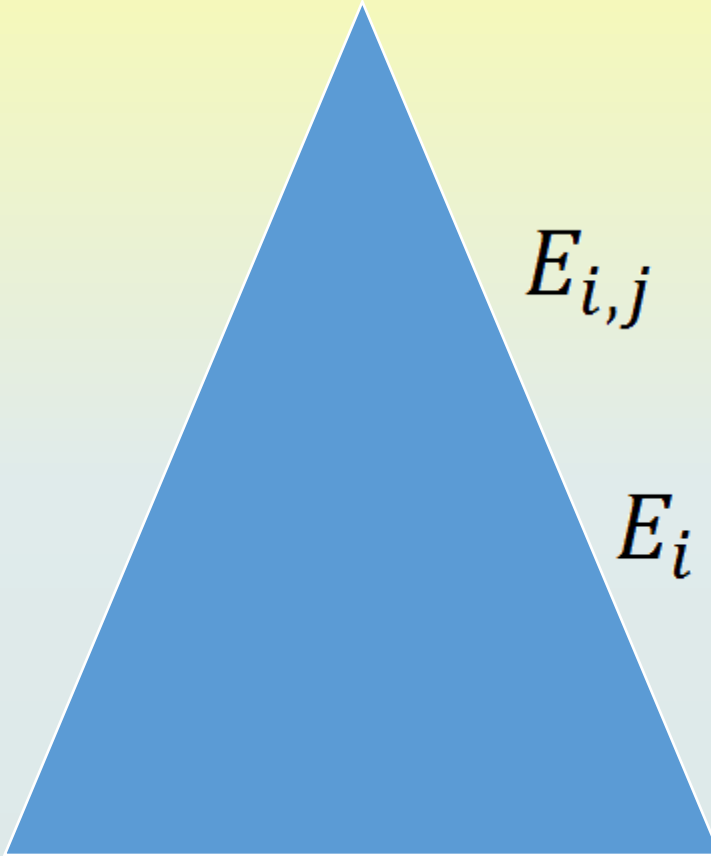
Simplified schematic framework



Methodology



Level of dis-aggregation



$$E_{i,j} = A_i \cdot S_{i,j} \cdot I_{i,j}$$

$$E_i = A_i \cdot \left(\sum_j S_{i,j} \cdot I_{i,j} \right)$$

A = Activity

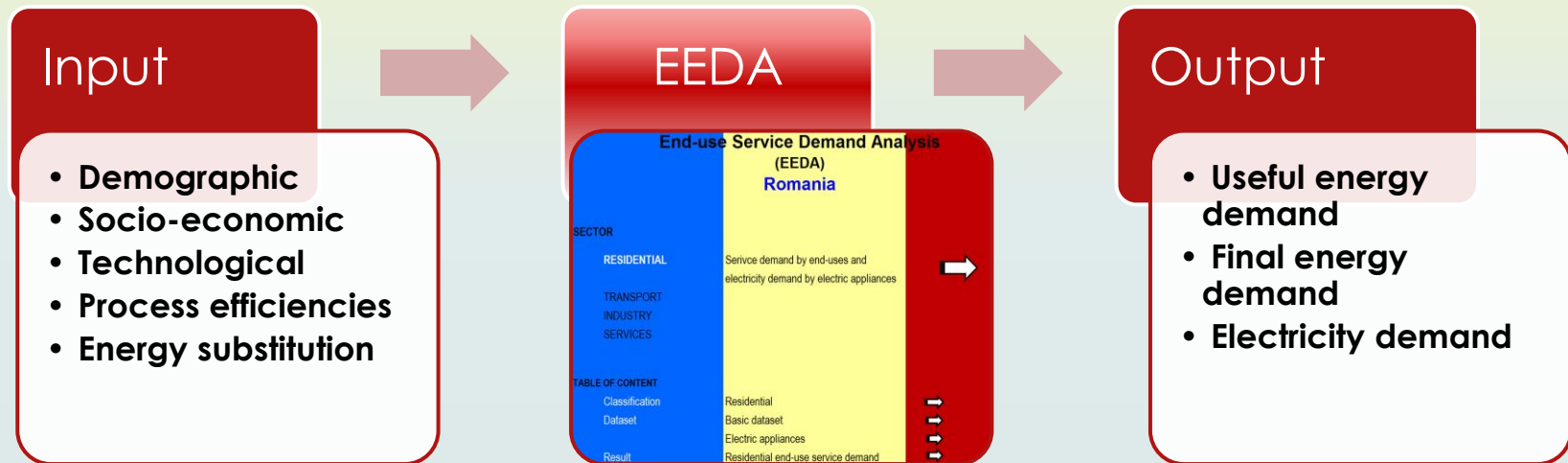
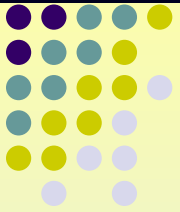
S = Structure

I = Energy intensity

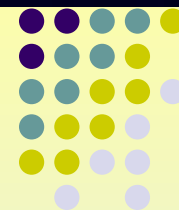
i = sector

j = sub-sector or end-use

Input requirements and output of the model



Methodology – Typology of sector/sub-sector and energy end-use services



Residential

- **Cooking**
- **Lighting**
- **Water heating**
- **Space heating**
- **Electric appliances**
 - Refrigerator
 - Air conditioner
 - Fan
 - Washing machine
 - Dish washer
 - Vacuum cleaner
 - Microwave oven
 - TV
 - Stereo
 - Computer
 - Iron
 - Other appliances



Services

- **Cooking**
- **Lighting**
- **Water heating**
- **Space heating**
- **Space cooling**
- **Electric appliances**
 - Refrigerator
 - Air conditioner
 - Fan
 - Washing machine
 - Dish washer
 - Vacuum cleaner
 - Microwave oven
 - TV
 - Computer
 - Other appliances



Industry

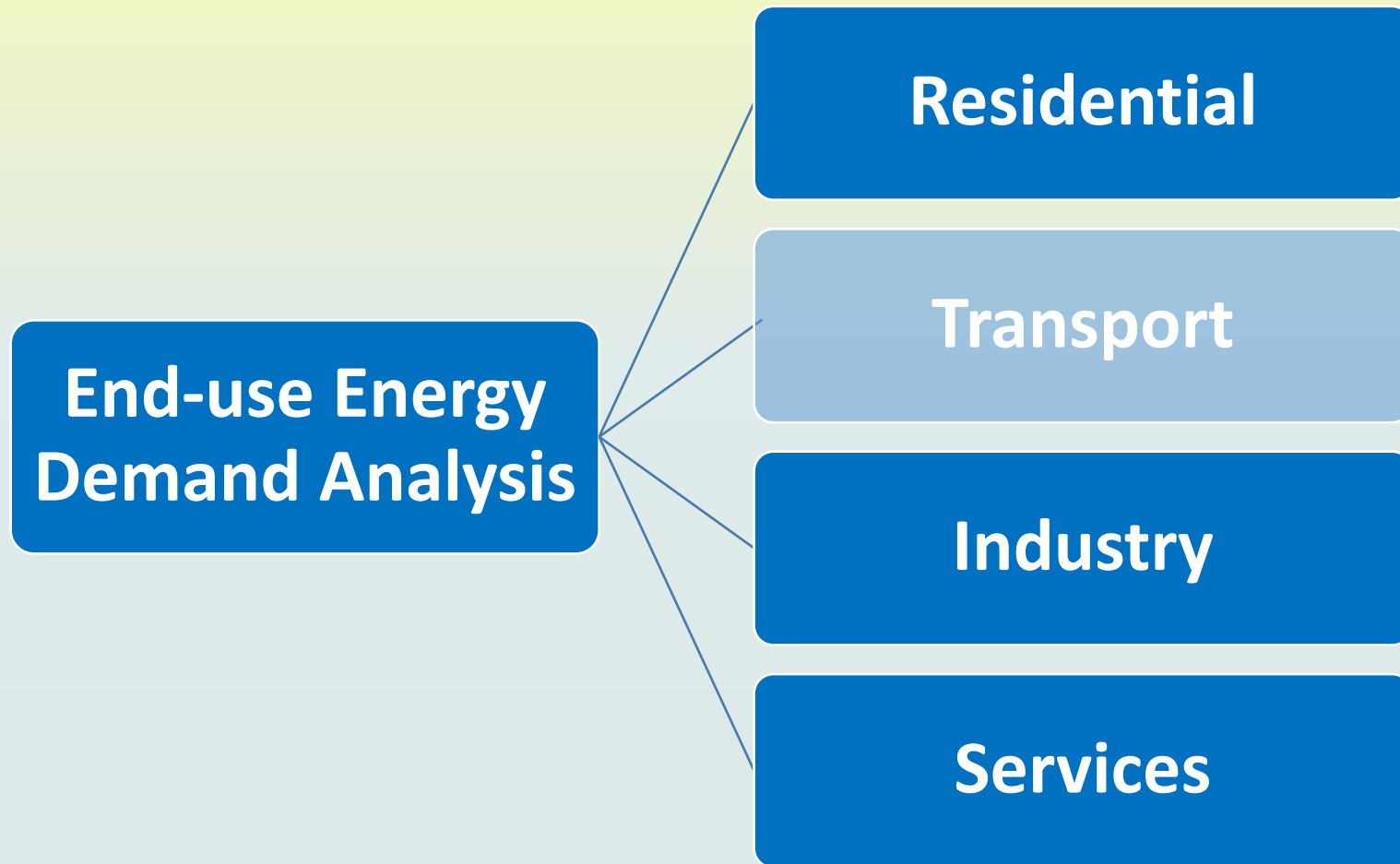
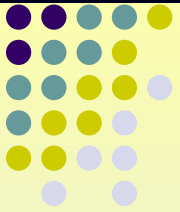
- **Agriculture**
- **Mining & quarrying**
- **Construction**
- **Manufacturing**
 - Iron & steel
 - Non-ferrous metals
 - Chemical & petrochemical
 - Non-metallic minerals
 - Food & tobacco
 - Textile & leather
 - Paper, pulp & print
 - Transport equipment
 - Machinery
 - Wood & wood products
 - Other manufacture



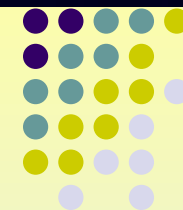
Transportation

- **Road**
 - Passenger
 - Freight
- **Railways**
 - Passenger
 - Freight
- **Air**
 - Passenger
 - Freight
- **Waterways**
 - Passenger
 - Freight
- **Pipelines**
 - Freight

End-use Energy Demand Analysis



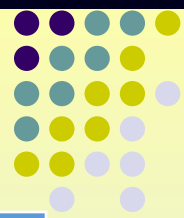
Residential



Sub-sector	A	S	I or (UEC)
Space heating	Floor area	Floor area/ dwelling	kgoe/floor area
Water heating	Population	Population/ dwelling	kgoe/person
Cooking	Population	Population/ dwelling	kgoe/person
Lighting	Floor area	Floor area/ dwelling	lumen/floor area
Electric appliances	No. of unit	Ownership/ dwelling	kWh/unit



Services

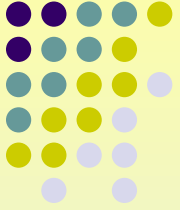


Sub-sector	A	S	I or (UEC)
Space heating	Value added (VA) Floor area	Share of VA Floor area/building	kgoe/floor area
Space cooling	Value added (VA) Floor area	Share of VA Floor area/building	kgoe/floor area
Lighting	Value added (VA) Floor area	Share of VA Floor area/building	lumen/floor area
Others	Value added (VA)	Share of VA	kgoe/floor area

Industry



Sub-sector	A	S	I or (UEC)
Iron and steel	Value added (VA)	Share of VA	kgoe/VA
Food and tobacco	Value added	Share of VA	kgoe/VA
Textile and leather	Value added	Share of VA	kgoe/VA
Construction	Value added	Share of VA	kgoe/VA



1. Overview of current energy demand situation in Romania

2. **End-use energy demand projection**

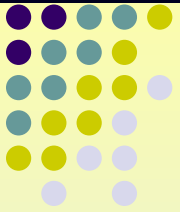
2.1 Methodology

2.2 Data and assumptions

2.3 Key results

3. Concluding remarks

Data type and Sources



❑ Three categories:

- ❑ Energy data
- ❑ Socio-economic and demographic data
- ❑ Technological data

❑ Main data sources:

- ❑ National Institute of Statistics (INS), Romania
- ❑ Macro model (population, GDP and GVA)
- ❑ Ministry of Environment and Climate Change (MMSC), Romania
- ❑ Eurostat and JRC Scientific and policy reports, EU
- ❑ Buildings Performance Institute Europe (BPIE), Brussels
- ❑ The World Development Indicators, World Bank

Energy balance

What are the available energy sources?

Supply

Transformation and energy industries own use

Non-energy use

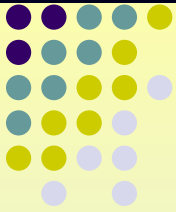
What is the total energy required? (Demand)

End-use energy sectors

Statistical differences

ROMANIA						
Year : 2010						
	Total all products	Hard coal	Coke	Lignite	Peat	Brown coal briquettes
	1000 toe					
Primary production	27,705	2		5,901	1	
Recovered products	67	0	0	0	0	0
Imports	11,711	519	736	21	8	0
Stock change	163	-6	-50	-74	0	0
Exports	3,975	46	3	0	0	0
Bunkers	16	0	0	0	0	0
Direct use	0					
Gross inland consumption	35,655	469	683	5,848	8	0
Transformation input	23,689	225	223	5,619	0	0
Main act. thermal power stations	7,782	183	0	5,227	0	0
Autoprod. thermal power stations	645	39	0	388	0	0
Nuclear power stations	2,968					
Briquetting plants	0	0	0	0	0	0
Coke-oven plants	2	2	0	0	0	0
Blast-furnace plants	223	0	223	0	0	0
Gas works	0	0	0	0	0	0
Refineries	11,350					
District heating plants	689	0	0	3	0	0
Transformation output	17,394		0			0
Main act. thermal power stations	4,071					
Autoprod. thermal power stations	286					
Nuclear power stations	999					
Briquetting plants	0					
Coke-oven plants	0		0			
Blast-furnace plants	223					
Gas works	0		0			
Refineries	11,327					
District heating plants	487					
Exchanges and transfers, returns	358					
Interproduct transfers	0					
Products transferred	0					
Returns from petrochem. industry	358					
Consumption of the energy branch	3,291	11	0	79	0	0
Distribution losses	1,425	5	1	2	0	0
Available for final consumption	25,002	229	458	147	8	0
Final non-energy consumption	2,308	0	0	0	0	0
Industry	1,681					
of which: chemical/petrochemical	1,064	0	0	0	0	0
Final energy consumption	22,509	180	599	152	8	0
Industry	6,909	180	599	142	0	0
Iron & steel industry	2,023	138	598	0	0	0
Non-ferrous metal industry	0	0	0	0	0	0
Chemical industry	2,056	37	0	90	0	0
Glass, pottery & building mat. industry	543	4	1	51	0	0
Ore-extraction industry	48	0	0	0	0	0
Food, drink & tobacco industry	557	0	1	0	0	0
Textile, leather & clothing industry	181	0	0	0	0	0
Paper and printing	165	0	0	0	0	0
Transport equipment	114	0	0	0	0	0
Machinery	365	0	0	0	0	0
Wood and wood products	263	0	0	0	0	0
Construction	408	0	0	0	0	0
Non specified (Other)						
Transport	5,013	0	0	0	0	0
Railways	221	0	0	0	0	0
Road transport	4,404					
International aviation	166					
Domestic aviation	111					
Domestic navigation	59	0	0	0	0	0
Other sectors	10,588	0	0	10	8	0
Households	8,102	0	0	10	0	0
Services	1,881	0	0	1	0	0
Agriculture/Forestry	391	0	0	0	8	0
Fishing	0	0	0	0	0	0
Statistical difference	186	50	-141	-5	0	0

Source: EUROSTAT.



Total final energy consumption =

Eurostat / IEA

❖ Non-energy use +

❖ Final energy consumption (FEC) +

- ❑ Industry (13 branches)
- ❑ Transport (6 modes)
- ❑ Households
- ❑ Services
- ❑ Agriculture
- ❑ Fishing

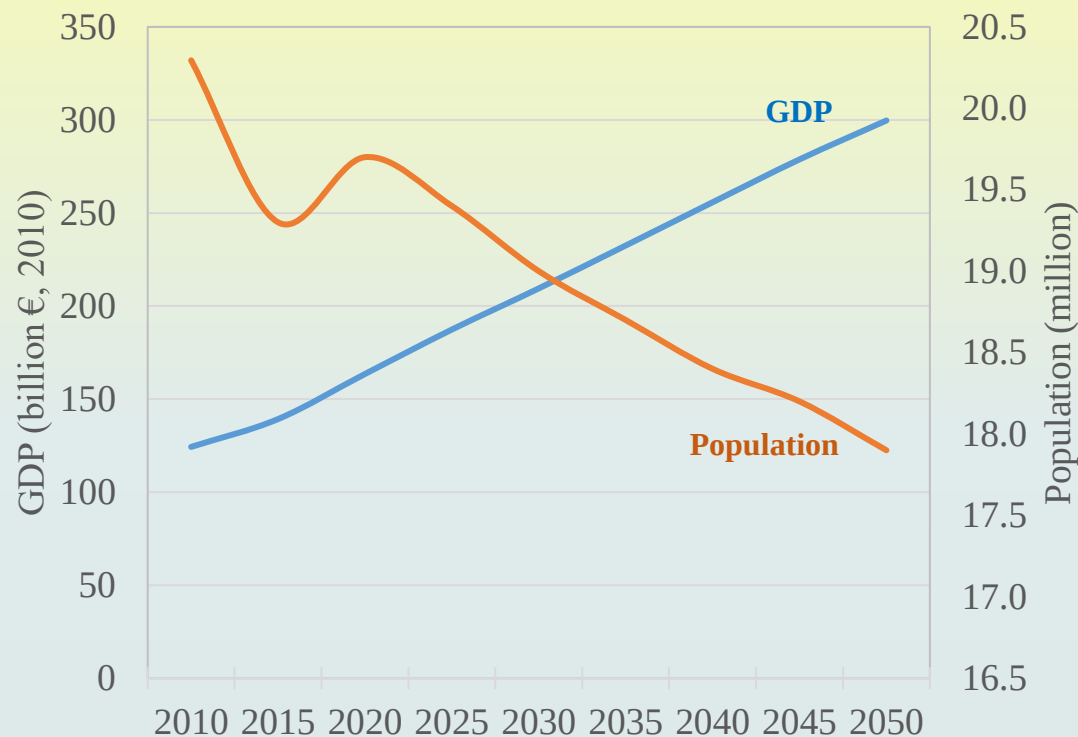
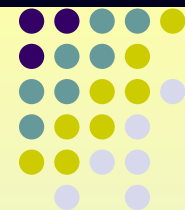
❖ Statistical differences +

- ❖ Iron & steel industry
- ❖ Non-ferrous metal industry
- ❖ Chemical industry
- ❖ Glass, pottery & building mat. industry
- ❖ Ore-extraction industry
- ❖ Food, drink & tobacco industry
- ❖ Textile, leather & clothing industry
- ❖ Paper and printing
- ❖ Transport equipment
- ❖ Machinery
- ❖ Wood and wood products
- ❖ Construction
- ❖ Non specified (Other)

- ❖ Railways
- ❖ Road transport
- ❖ International aviation
- ❖ Domestic aviation
- ❖ Domestic navigation
- ❖ Pipeline transport

No breakdown available

Data and assumptions – Socio-economic data



Source: Macro model.

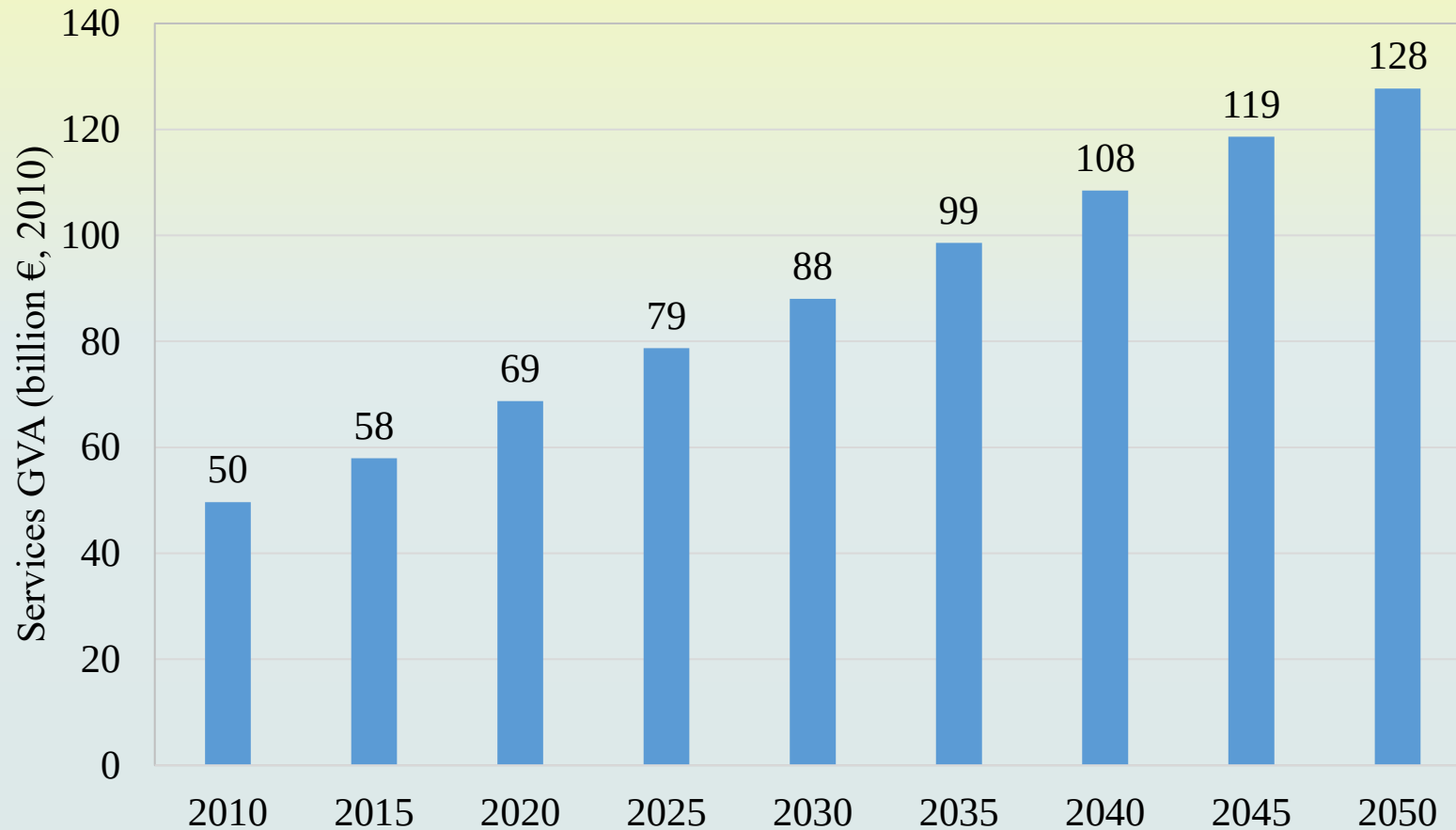
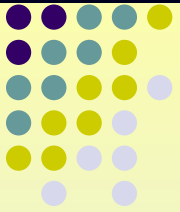
Population:

- ❑ AAGR: -0.3% (2010-2050)
- ❑ 20.3 million in 2010 to 17.9 million in 2050

GDP (real):

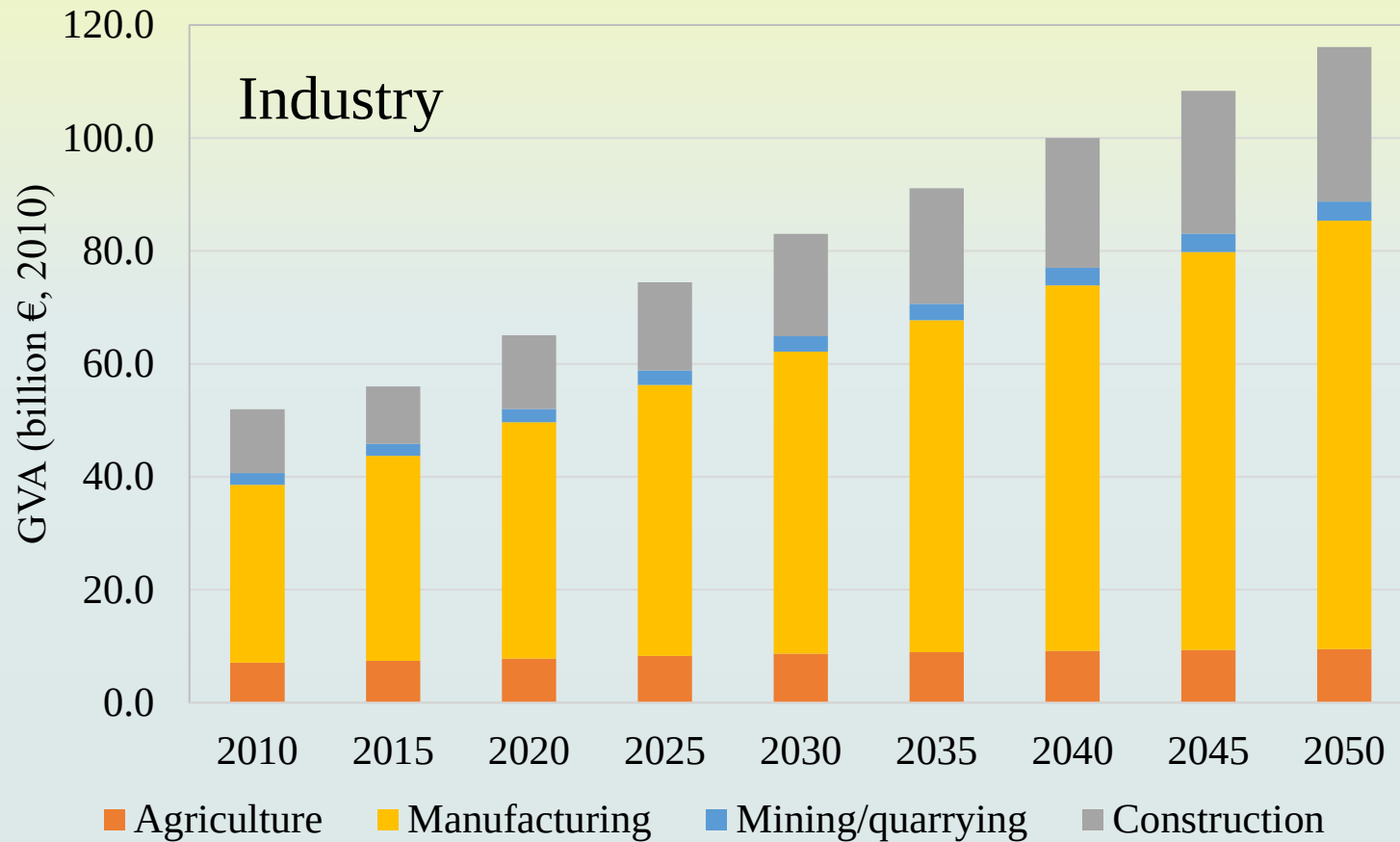
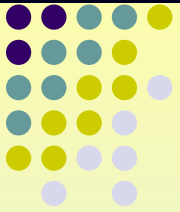
- ❑ AAGR: 2.2% (2010-2050)
- ❑ 2.4 times increase from 2010 to 2050

Data and assumptions – Economic drivers (Services sector)



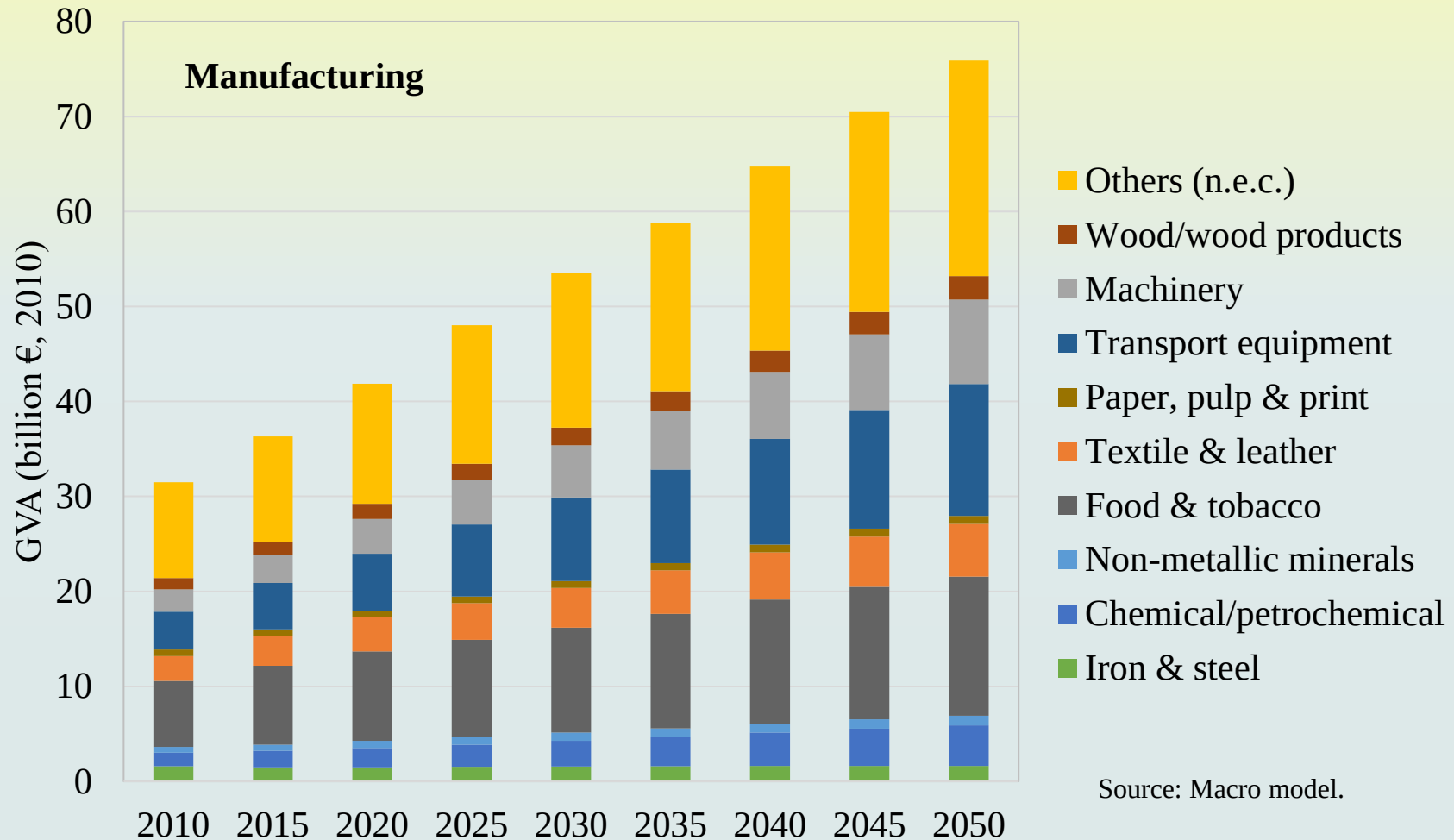
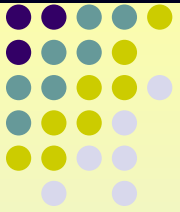
Source: Macro model.

Data and assumptions – Economic drivers (Industry)

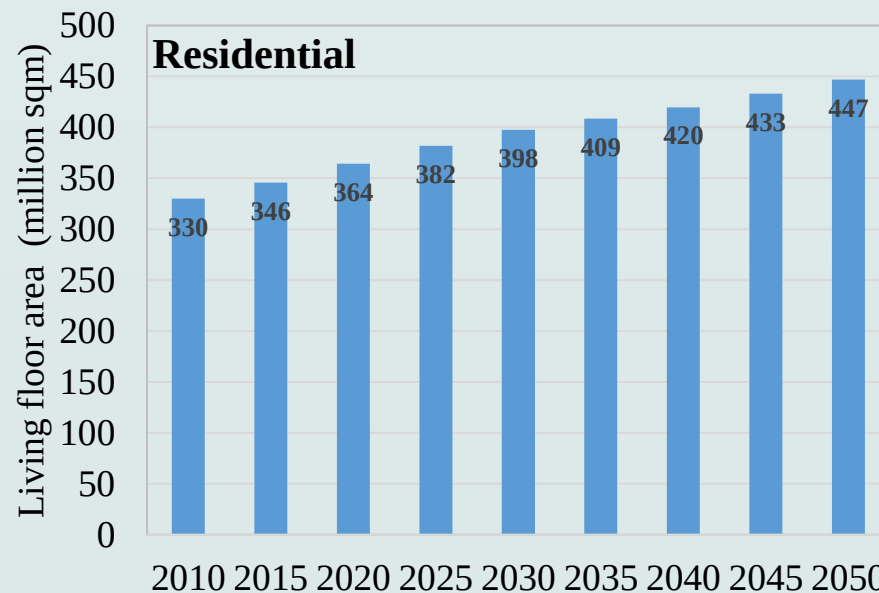
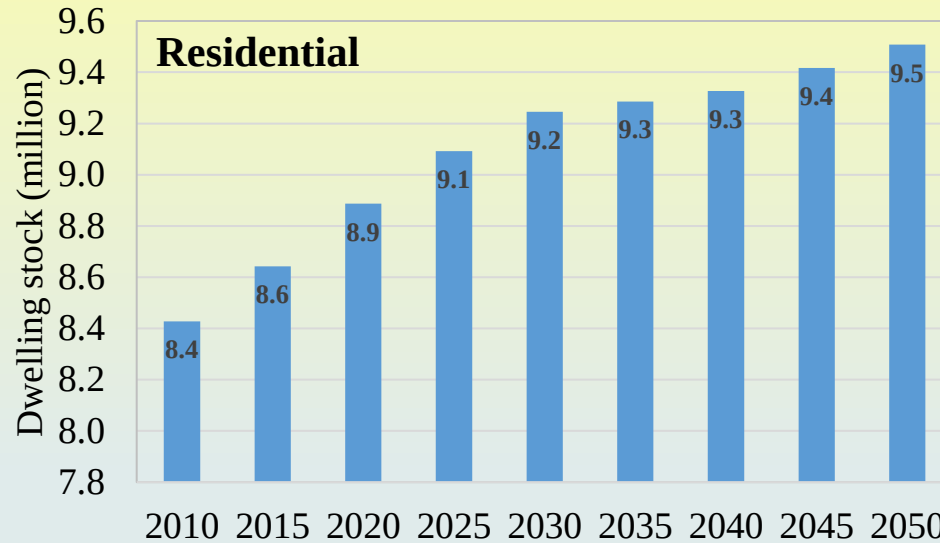
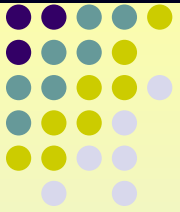


Source: Macro model.

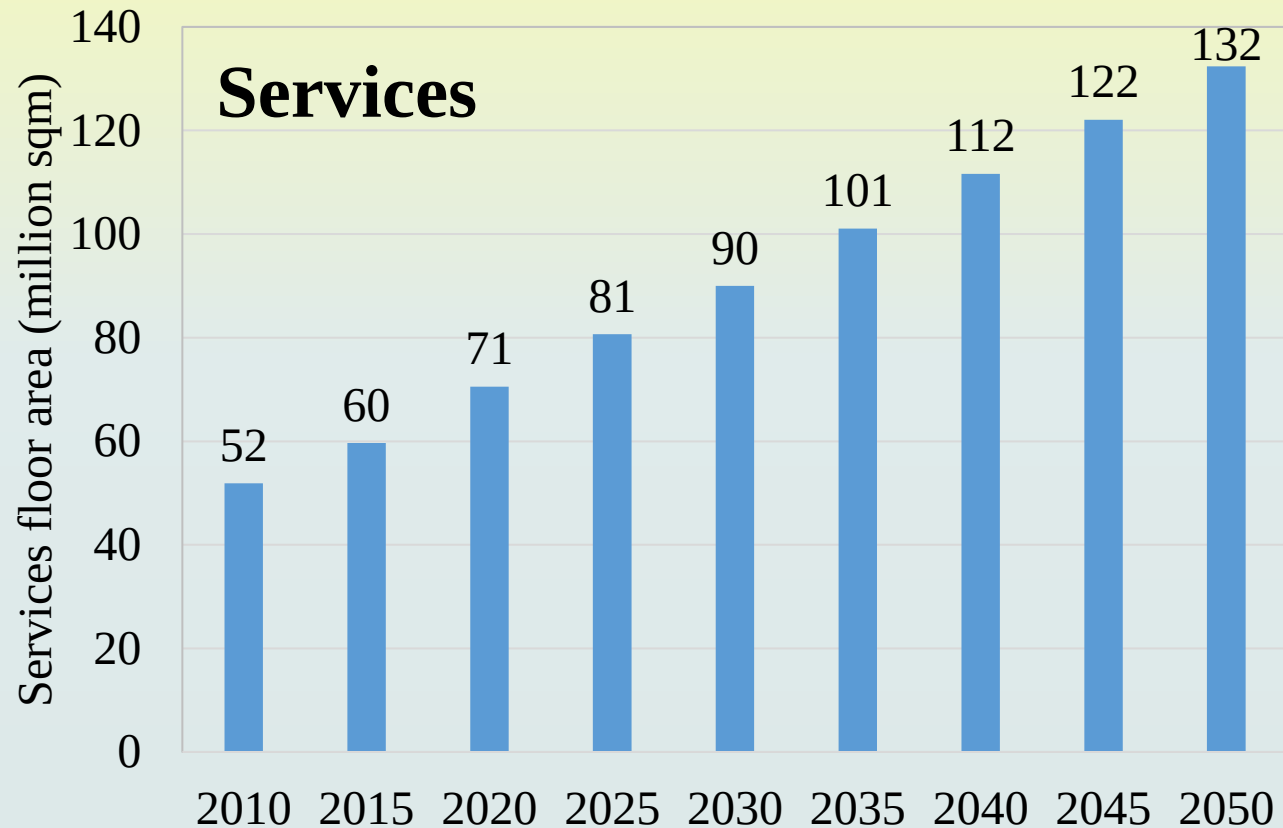
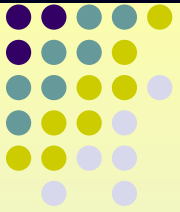
Data and assumptions – Economic drivers (Manufacturing)



Data and assumptions – Technological drivers



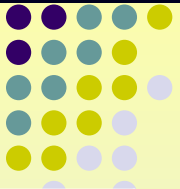
Data and assumptions – Technological drivers



Residential sector

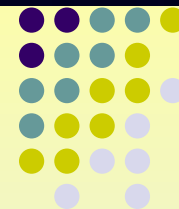
Drivers / parameters	Historical data	Future (data source and assumptions)
Average HDD (1980-09)	EUROSTAT (2014)	Assumed same value (i.e., 3092) for future years.
No. of households (Dwelling stock)	INS	$DWT(t+1) = DWT(t) \cdot (1 + r / 100)^n \cdot (1 - DRDW / 100)$ where <i>DWT</i> is dwelling stock (no. of households), <i>r</i> is annual dwelling growth rate and <i>DRDW</i> is dwelling demolition rate. <i>r</i> is assumed at 0.7% per year for 2015-2050. <i>DRDW</i> for 2015-2030 is based on MOE (2013) and 2035-250 is authors' assumption.
Floor area	INS	Calculated
Floor area/dwelling	Calculated	$APDW(t+1) = APDW(t) \cdot (1 + r)^n$ where <i>APDW</i> is dwelling area/dwelling and <i>r</i> is annual APDW growth rate. <i>r</i> is assumed at 0.5% in every 5 yr for 2015-2040 and at 0.4% every 5 yr for 2040-2050.
Population	INS	Macro model
Household size	Calculated	Calculated
Dwelling occupancy rate	BPIE (2014)	
Appliances ownership	INS	Appliances ownership for 2050 is authors' assumption. Figures for 2015-2045 is interpolation of 2010 (actual) and 2050 values.
Specific energy consumption (SEC) for end-uses	Pardo et al. (2012); Sanga et al. (2005)	SEC for lighting is based on 150 lux with 5 hrs per day and 25/1500 W/lm (based on 25 W CFL).
Unit energy consumption (UEC) for appliances and AEEI	EU (2014)	Data for 2015-2030 is from EU (2014) based on autonomous scenario and 2035-2050 is authors' assumption. AEEI is autonomous energy efficiency improvement.

Services sector



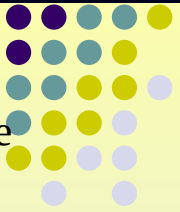
Drivers / parameters	Historical (data source)	Future (data source and assumptions)
Floor area by services type	BPIE (2014) Floor area for 2010 is assumed based on 2009 value (i.e., 67.2 m sqm) and average growth of service sector GVA of 2.3% between 2009 and 2010. However, the share of sub-sectors in total floor area for 2010 is assumed same as that of 2009.	Future growth rate of services floor area and demolition rate for 2010-2030 are based on MOE (2013) ; and for 2030-2050 are based on authors' assumption.
SEC for all end-uses except others	Pardo et al. (2012)	SEC for space heating and cooling, and water heating is calculated in useful energy per unit of floor area. SEC for lighting is considered same as residential sector.
SEC for others end-use	Atanasiu et al. (2012)	SEC for others (in final energy) for 2015-2050 is based on AEEI of 1% in every 5 yr between 2010-2030 and 0.5% in every 5 yr between 2030-2050.

Industry sector

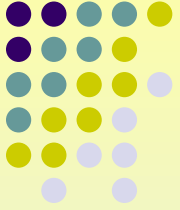


Drivers / parameters	Historical (data source)	Future (data source and assumptions)
GDP, GVA and growth rate	EUROSTAT (2014)	Macro model
Energy intensity for thermal energy use by end-uses	Pardo et al. (2012)	
Energy intensity for motor fuels and electricity	EUROSTAT (2014)	Motor fuels are not used for thermal energy requirements. Electricity for thermal energy use is assumed 5% of total electricity use for agriculture and manufacturing, and 0% for mining and quarrying, and construction. AEEI is assumed 0.5% in every 5 yr during 2010-2050 for motor fuels. For non-thermal electricity use, it is assumed at 1% in every 5 yr during 2010-2050. About 60% of total electricity use in industry are used by electric motor systems based on Bertoldi and Atanasiu (2009).

Selected references of data sources

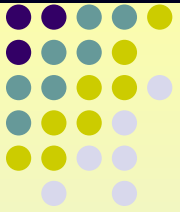


- Atanasiu, B., M. Economidou and J. Maio (2012). The Challenges, Dynamics and Activities in the Building Sector and its Energy Demand in Romania. Buildings Performance Institute Europe (BPIE) and Enerdata, Brussels.
- Bertoldi, P. and B. Atanasiu (2009). Electricity Consumption and Efficiency Trends in European Union. Status Report 2009, Joint Research Centre, Institute for Energy, European Union (EU), Ispra (VA), Italy.
- BPIE (2014). Renovating Romania: Strategy for the Energy Renovation of Romania's Building Stock. BPIE, Brussels.
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- EU (2014). Database on Energy Savings Potentials. European Union (EU).
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- INS
1. Tempo-Online Time series (Web link: <https://statistici.insse.ro/shop/?lang=en>)
 2. Romania in Figures (several issues) (Web link: http://www.insse.ro/cms/files/publicatii/Romania_in_Figures_2014.pdf)
 3. Statistical Yearbook 2012 (CD version 2014)
- MOE (2013). Romania's 6th National Communication on Climate Change and First Biennial Report. Ministry of Environment and Climate Change, Bucharest.
- Pardo, N., K. Vatopoulos, A. Krook-Riekkola, J. A. Moya and A. Perez (2012). Heat and Cooling Demand and Market Perspective. JRC 70962, Joint Research Center, Institute for Energy and Transport, EU, Petten, The Netherlands.
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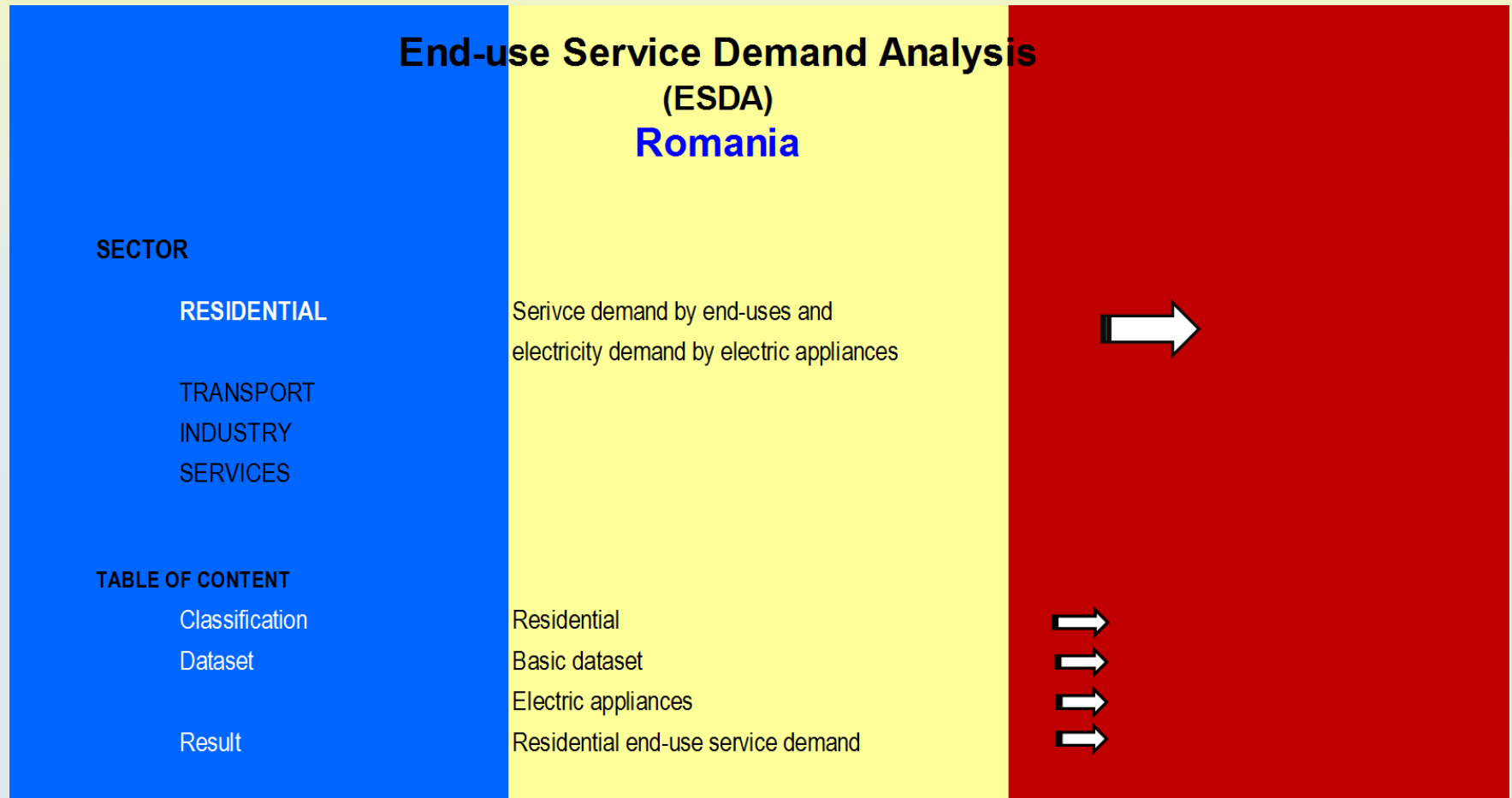


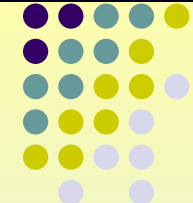
1. Overview of current energy demand situation in Romania
2. **End-use energy demand projection**
 - 2.1 Methodology
 - 2.2 Data and assumptions
 - 2.3 **Key results**
3. Concluding remarks

End-use Service Demand Analysis Model



Residential Sector





Residential

**Space
heating**

**Water
heating**

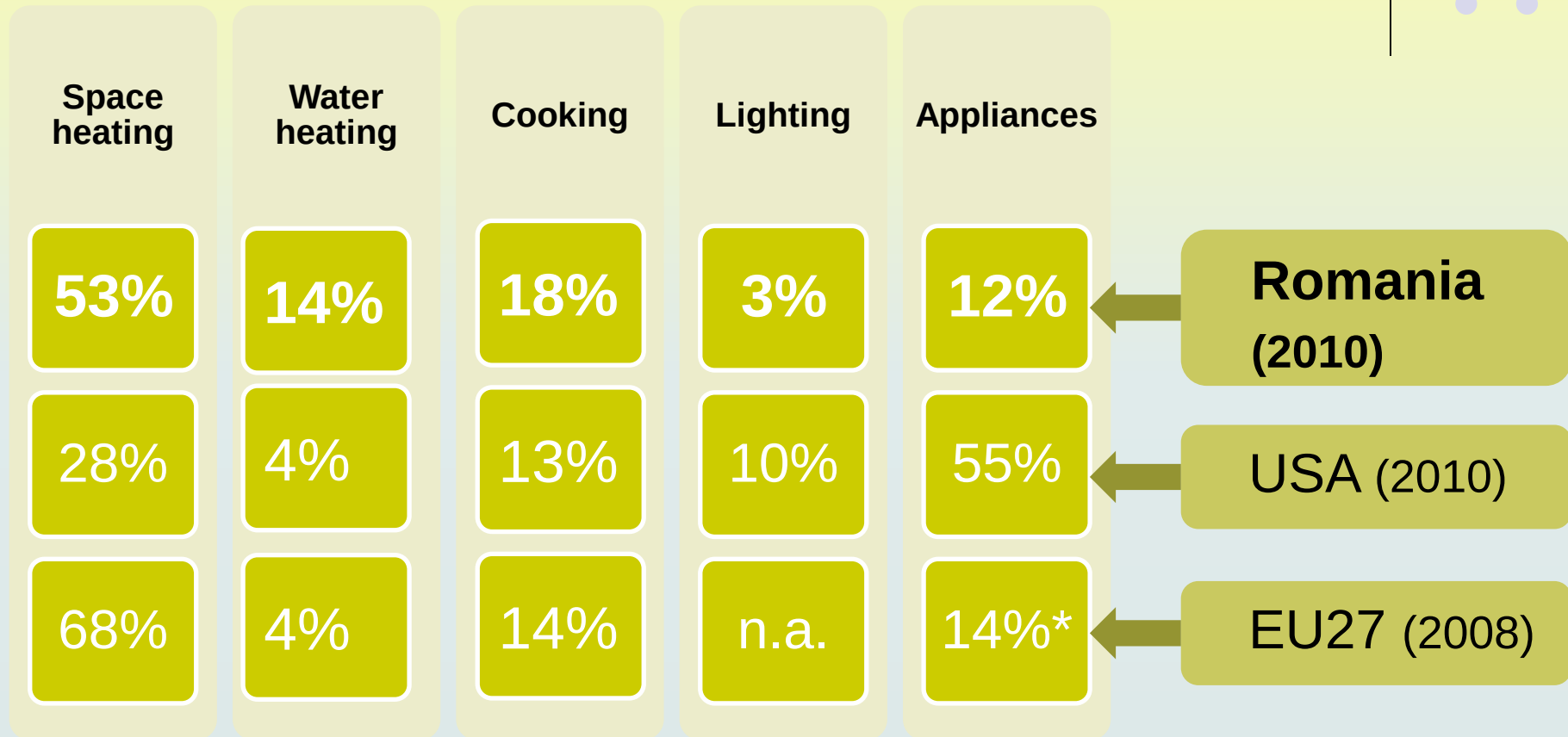
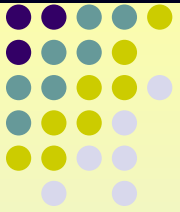
Cooking

Lighting

**Electric
appliances:**

- Refrigerator
- Washing m/c
- AC
- TV
- Computer
- Others

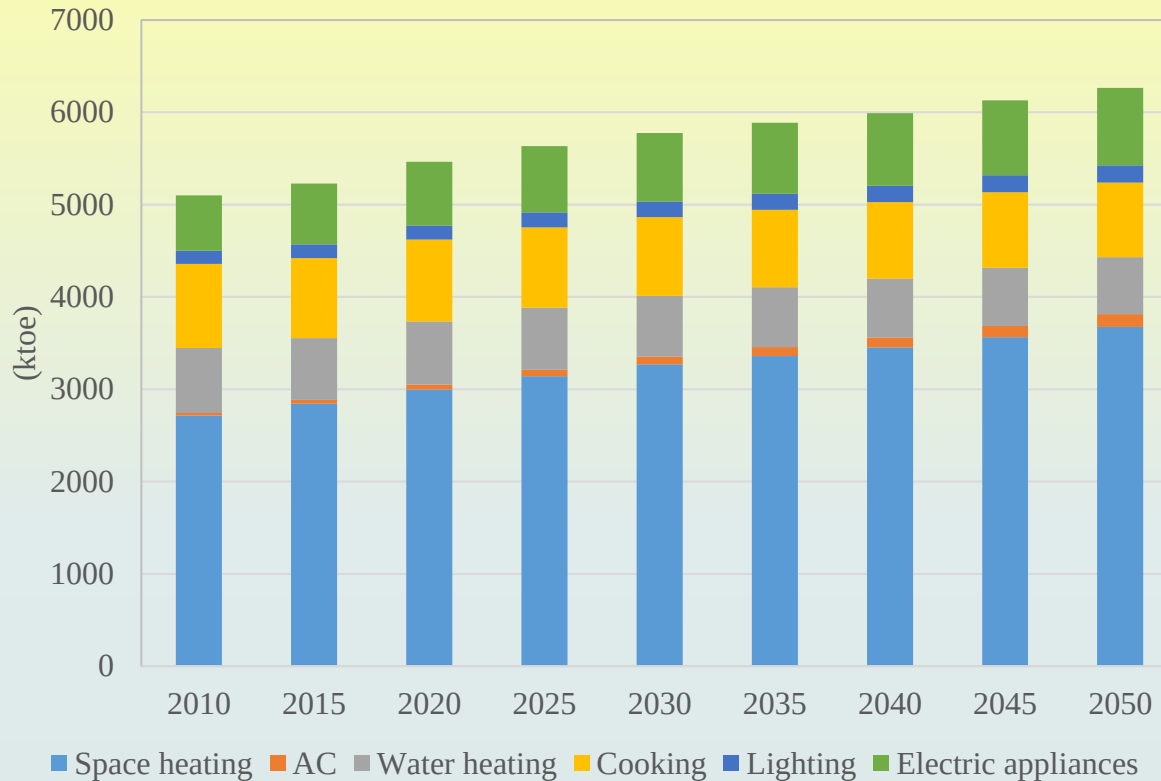
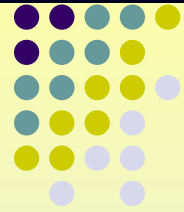
Residential end-use patterns



Note: * Includes lighting.

Sources: Romania (ESDA model) USA (EIA, 2012); EU 27 (EEA).

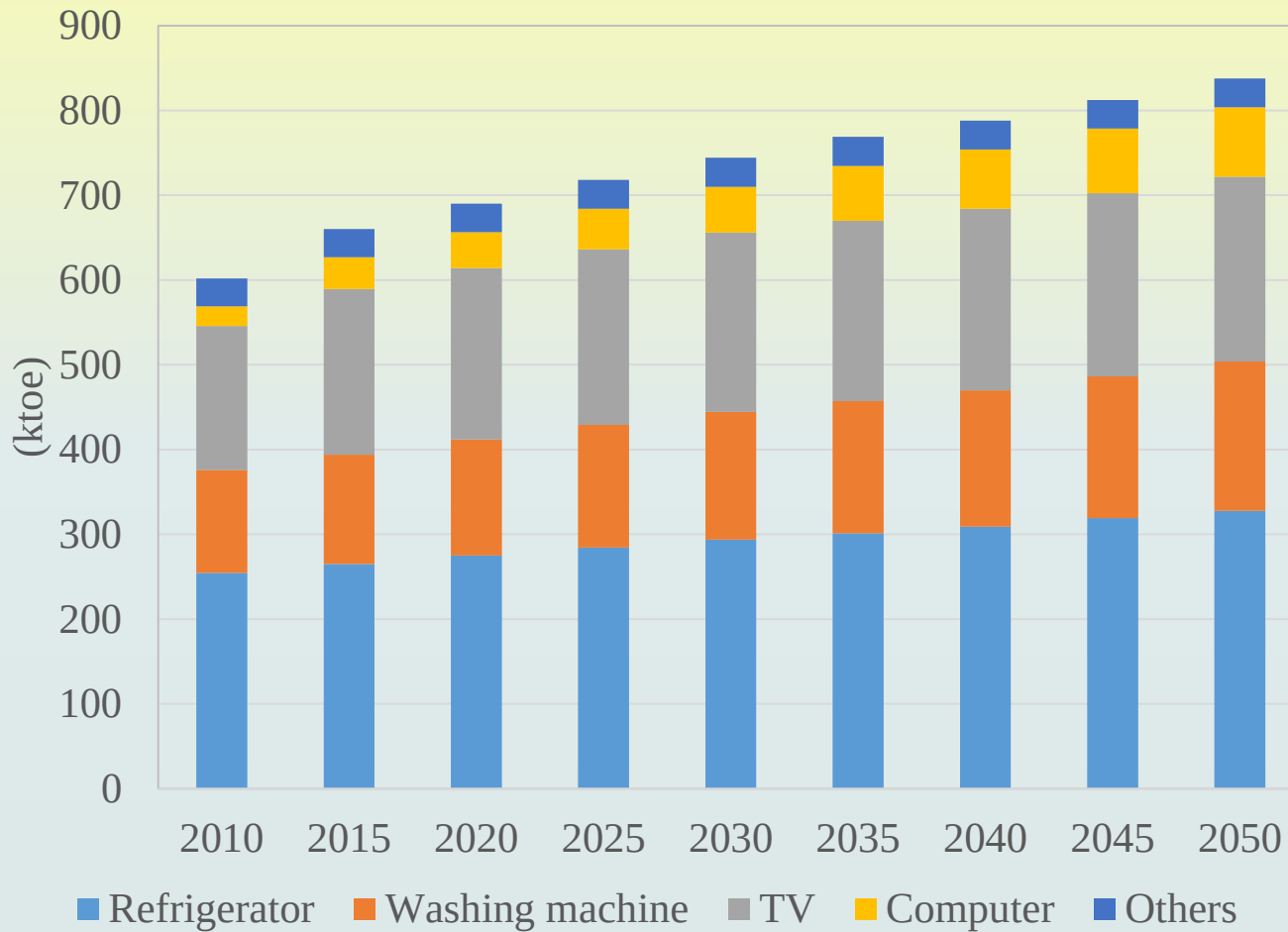
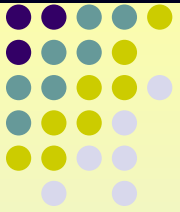
Residential sector by end-use (ktoe)



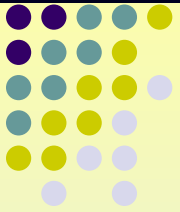
- ❑ Residential sector is Romania's largest energy consumer in 2010.
- ❑ Space heating is projected to account for most of energy demand in residential sector.

Residential end-use	2010	2015	2020	2025	2030	2035	2040	2045	2050
Space heating (climate corrected)	53%	54%	55%	56%	57%	57%	58%	58%	59%
Air conditioning	<1%	1%	1%	1%	1%	2%	2%	2%	2%
Water heating	14%	13%	12%	12%	11%	11%	11%	10%	10%
Cooking	18%	17%	16%	15%	15%	14%	14%	13%	13%
Lighting	3%	3%	3%	3%	3%	3%	3%	3%	3%
Electric appliances	12%	13%	13%	13%	13%	13%	13%	13%	13%

Residential sector electric appliances (ktoe)

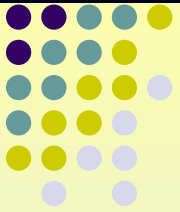


Overall residential efficiency in 2010



$$\frac{\text{Useful energy Demand (5101)}}{\text{Final energy Demand (8102)}} = \eta (63\%)$$

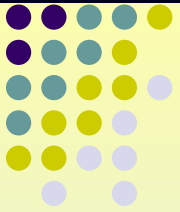
Comparison (Residential)



		2015	2020	2025	2030	2035	2040	2045	2050
PRIMES	ktoe	8558	8575	8710	8667	8995	9210	9369	9515
Overall efficiency	%	61	64	65	67	65	65	65	66

MMSC	ktoe	8489	9391	9702	10048
Overall efficiency	%	62	58	58	57

End-use Service Demand Analysis Model



Services Sector

End-use Service Demand Analysis (ESDA) Romania

SECTOR

RESIDENTIAL
TRANSPORT
INDUSTRY
SERVICES

TABLE OF CONTENT

Classification
Dataset
Result

Service demand by end-uses

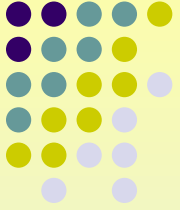


Services

Basic dataset

Services end-use service demand





Services

Space heating

- Office
- Hospital
- Hotel
- Sport facility
- Others

Space cooling

- Office
- Hospital
- Hotel
- Sport facility
- Others

Water heating

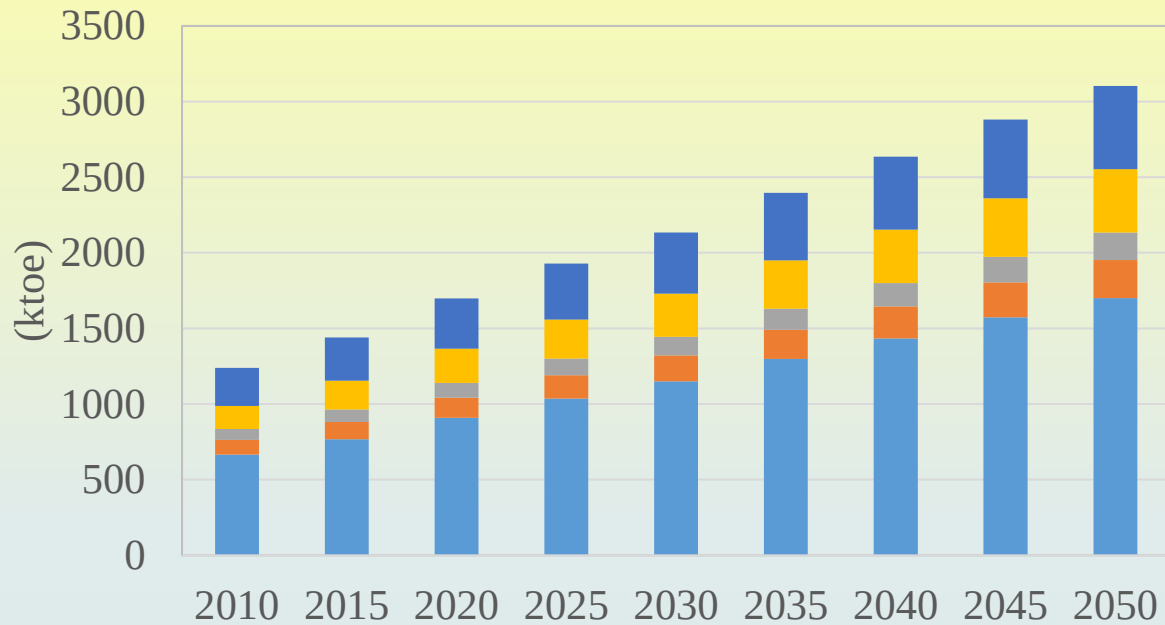
- Office
- Hospital
- Hotel
- Sport facility
- Others

Lighting

- Office
- Hospital
- Hotel
- Sport facility
- Others

Others

Services sector by end-use (ktoe)



■ Space heating ■ Space cooling ■ Water heating ■ Lighting ■ Others*

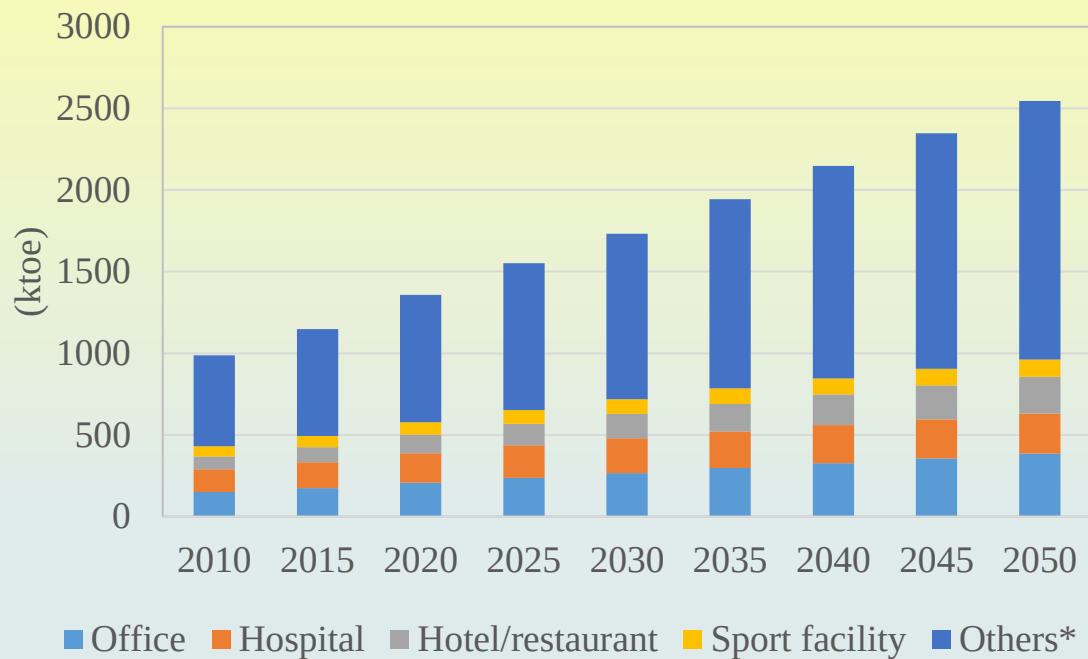
Services end-use	2010	2015	2020	2025	2030	2035	2040	2045	2050
Space heating	54%	53%	54%	54%	54%	54%	54%	55%	55%
Space cooling	8%	8%	8%	8%	8%	8%	8%	8%	8%
Water heating	6%	6%	6%	6%	6%	6%	6%	6%	6%
Lighting	12%	13%	13%	13%	13%	13%	13%	13%	13%
Others*	20%	20%	20%	19%	19%	19%	18%	18%	18%

□ Services sector is Romania's smallest energy consumer in 2010 but projected to grow at much higher rate than other sectors.

□ Space heating is projected to account for most of energy demand in services sector.

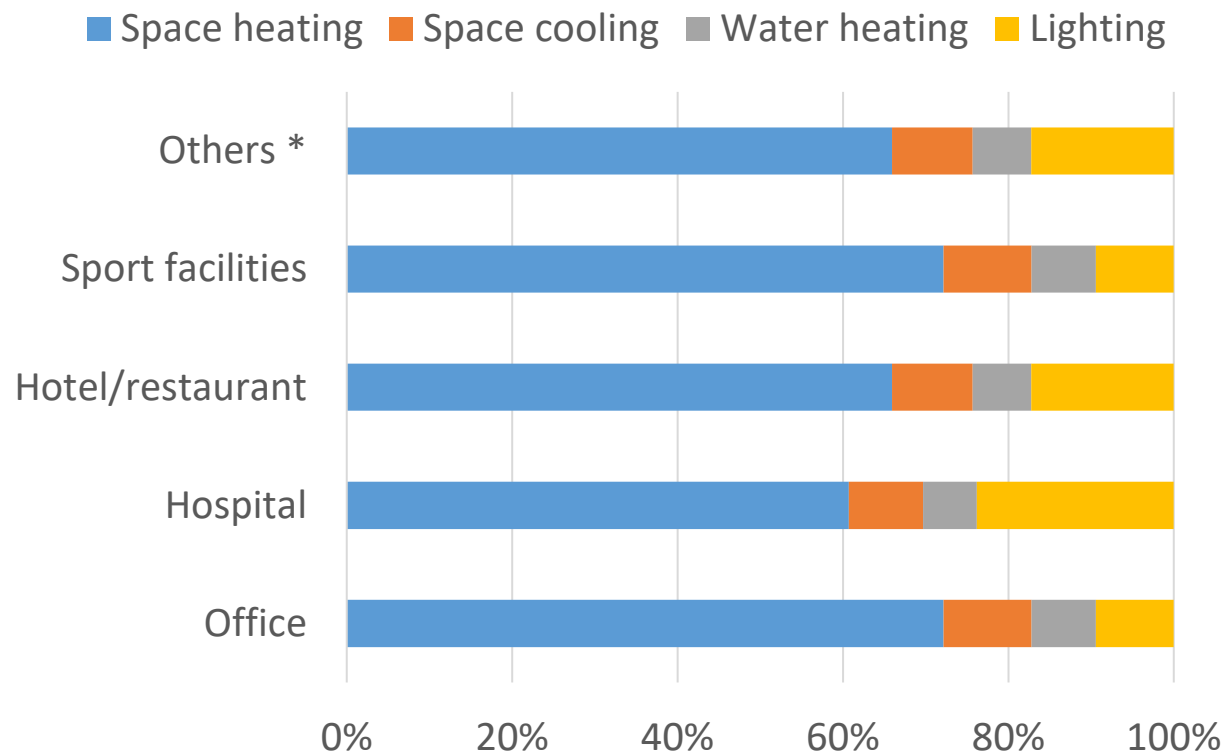
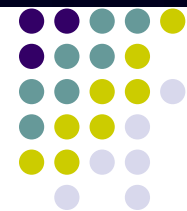
Note: * Include educational, wholesale/retail trade, and other non-residential buildings.

Services sector by sub-sector (ktoe)

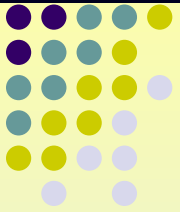


Services sub-sector	2010	2015	2020	2025	2030	2035	2040	2045	2050
Office	15%	15%	15%	15%	15%	15%	15%	15%	15%
Hospital	14%	14%	13%	13%	12%	12%	11%	10%	10%
Hotel/restaurant	8%	8%	8%	8%	9%	9%	9%	9%	9%
Sport facility	7%	6%	6%	6%	5%	5%	5%	4%	4%
Others*	56%	57%	57%	58%	58%	60%	61%	61%	62%

Services sector by sub-sector and by end-use in 2050

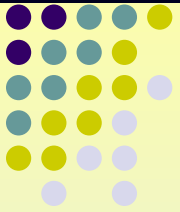


Overall services efficiency in 2010



$$\frac{\text{Useful energy Demand (1239)}}{\text{Final energy Demand (1881)}} = \eta (66\%)$$

End-use Service Demand Analysis Model



Industry Sector

End-use Service Demand Analysis (ESDA) Romania

SECTOR

RESIDENTIAL
TRANSPORT
INDUSTRY
SERVICES

TABLE OF CONTENT

Classification
Dataset
Result

Service demand by end-uses



Services

Basic dataset

Services end-use service demand



Industry

Agriculture

Iron and steel

Chemical and petro-chemical

Non-metallic minerals

Food and tobacco

Textile and leather

Paper, pulp and print

Transport equipment

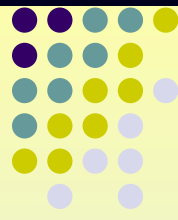
Machinery

Wood and wood products

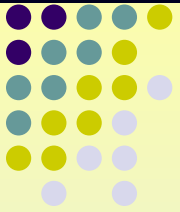
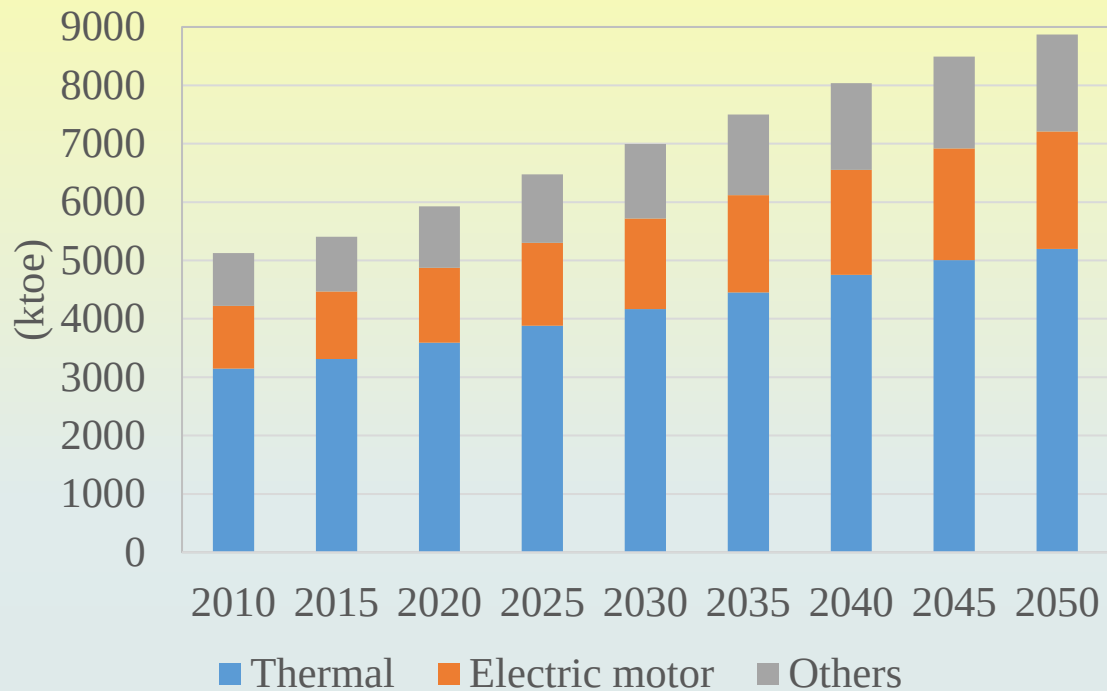
Other manufacturing

Mining and quarrying

Construction



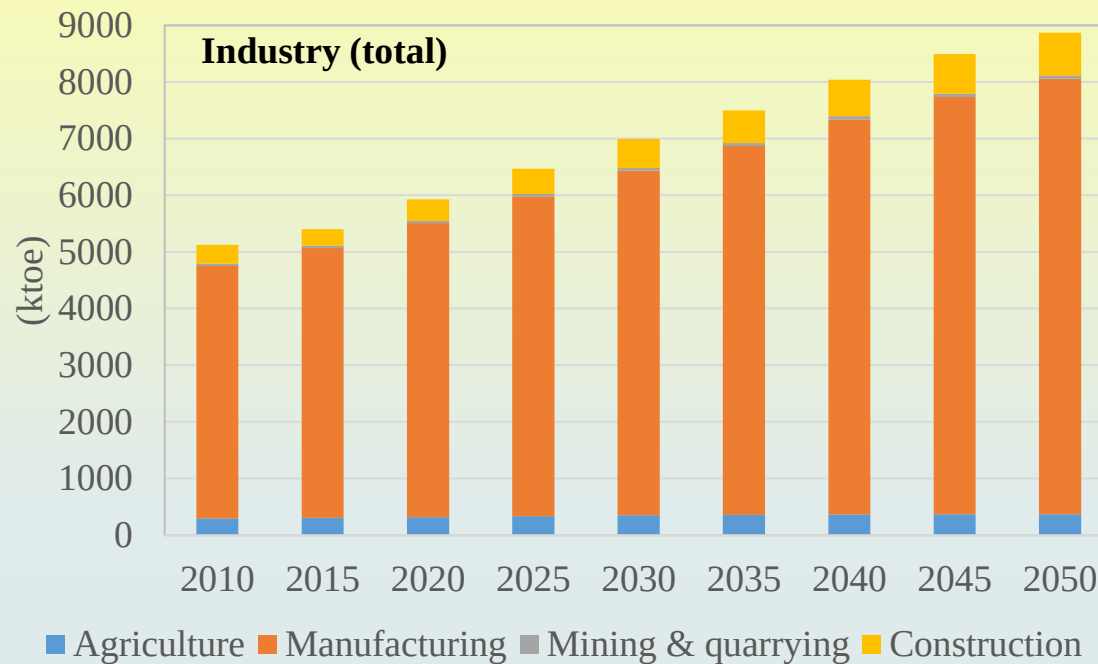
Industry sector by end-use (ktoe)



Note: Industry sector includes agriculture, mining and quarrying, and construction.

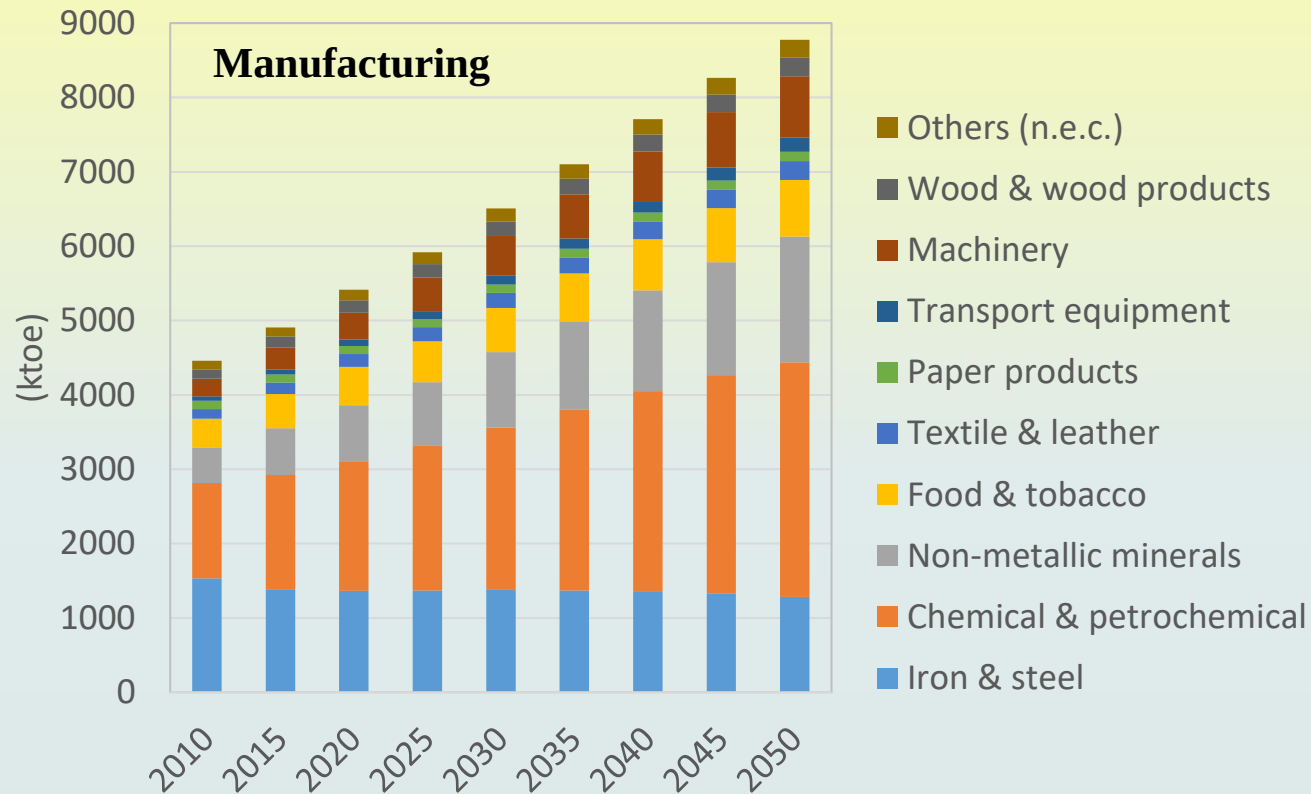
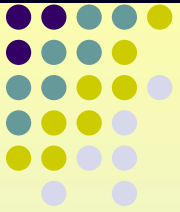
Industry end-use	2010	2015	2020	2025	2030	2035	2040	2045	2050
Thermal	61%	62%	61%	61%	60%	60%	60%	60%	60%
Electric motor	21%	21%	21%	22%	22%	22%	22%	22%	22%
Others	18%	17%	17%	18%	18%	18%	18%	18%	18%

Industry sector by sub-sector (ktoe)

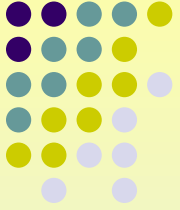


Industry sub-sector	2010	2015	2020	2025	2030	2035	2040	2045	2050
Agriculture	6%	6%	5%	5%	5%	5%	5%	4%	4%
Manufacturing	87%	88%	87%	87%	87%	87%	87%	87%	87%
Mining & quarrying	1%	1%	1%	1%	1%	1%	1%	1%	1%
Construction	6%	5%	6%	7%	7%	8%	8%	8%	9%

Manufacturing by sub-sector (ktoe)

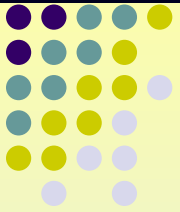


Manufacturing sub-sector	2010	2015	2020	2025	2030	2035	2040	2045	2050
Iron and steel	34%	28%	25%	23%	21%	19%	18%	16%	15%
Chemical/ petro-chemical	29%	31%	32%	33%	34%	34%	35%	36%	36%
Others	37%	40%	43%	44%	45%	46%	47%	48%	49%



1. Overview of current energy demand situation in Romania
2. End-use energy demand projection
 - 2.1 Methodology
 - 2.2 Data and assumptions
 - 2.3 Key results
3. **Concluding remarks**

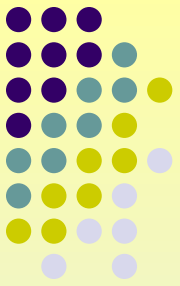
Concluding Remarks



- ☐ **Total Romanian end-use demand is projected to increase from 2010 to 2050.**
- ☐ **Most of these end-use energy demand is coming from residential and industrial sectors.**
- ☐ **Demand for space heating is projected to be the dominant end-use service in residential and services sectors during 2010-2050.**
- ☐ **Space heating represents largest opportunity to reduce energy demand in residential and services by increasing energy efficiency of heating equipment and buildings renovation, and by changing energy-mix.**
- ☐ **In industry sector, there is potential of reducing electricity demand by increasing energy efficiency of electric motor driven systems.**

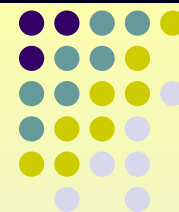


THANK YOU.



EXTRA SLIDES

Policy and measures (Romania)



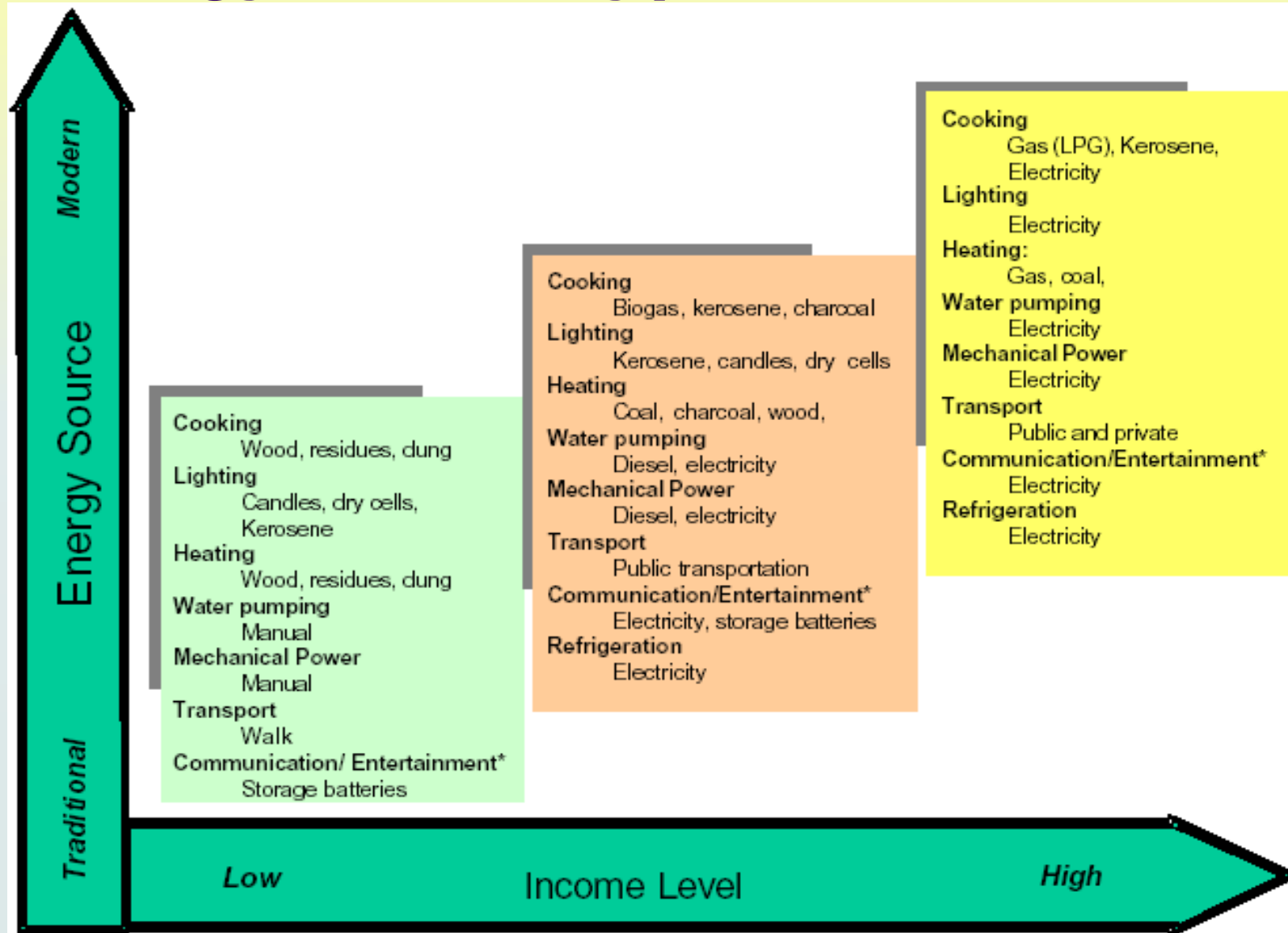
EU Directive	Policy/Measure
2006/32/EC	Energy end-use efficiency and energy services
2005/32/EC	Establish framework for the setting of eco-design requirements for energy-using products
2009/28/EC	promotion of the use of energy from renewable sources
2004/8/EC	promotion of cogeneration based on a useful heat demand in the internal energy market
2009/33/EC	promotion of clean and energy-efficient road transport vehicles
2009/443/EC	establishing the emission performance standards for new passenger cars
2010/40/EU	framework for the deployment of ITS in the field of road transport and for interfaces with other modes of transport

24% of the final energy consumption will be cover by renewable energy resources in 2020. (**National action plan for renewable energy**)

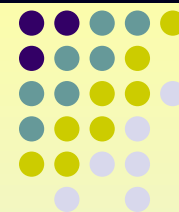
The **2nd National Action Plan on Energy Efficiency for 2011 – 2020** consider promotes measures for efficient use of energy (**20% reduction in year 2020**)

Implementation of **biofuels Directive** (Transport)

Energy ladder hypothesis

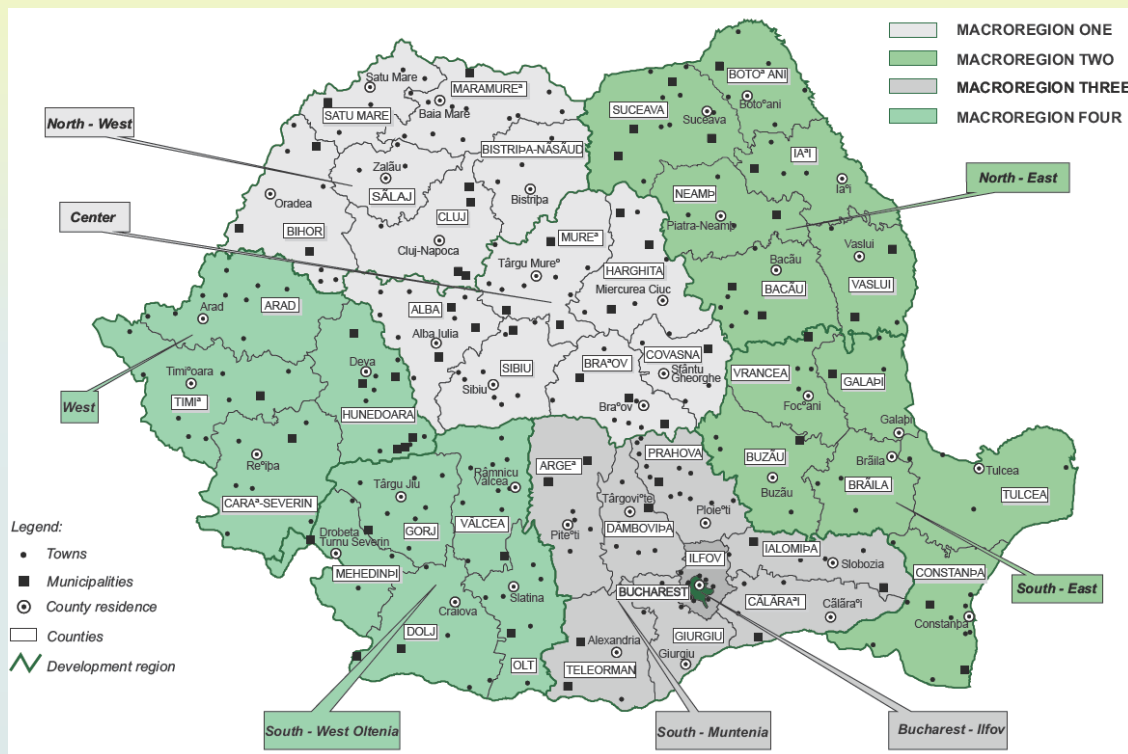
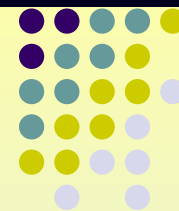


Key terms



- ❖ **Useful energy** is the energy available to the consumers after the last conversion made in the consumer energy conversion equipment, hence final energy consumption minus conversion losses
- ❖ **Final energy consumption** is defined by Eurostat as the energy supplied to the final consumer's door.
- ❖ **Energy Intensity**: Energy 'consumed' per unit of activity or output.
- ❖ **Heating Degree-Days** are the number of degrees per day that the daily average temperature is below a given temperature (18°C or 65°F). The given temperature is the point below which the consumer is assumed to use fuels for space heating and varies from country to country.
- ❖ **Drivers** are explanatory factors that have a direct physical relationship with energy use. In the residential sector it is population, number of households, number and size of dwellings, central/local heating and ownership of appliances.

Residential is heterogeneous

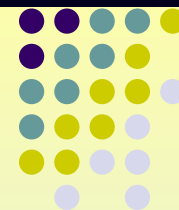


Depending on data availability, disaggregation is made using homogenous categories, e.g.:

- Urban/rural areas
- Macro/development region
- Income and socio-economic classes

Figure source: Romania in Figures (NIS, 2013)

Adjusted HDD calculation



$$\text{Actual energy for SH (current year)} \times \frac{\text{HDD (current year)}}{\text{Average HDD (period)}}$$

	2010
HDD (current year)	2773
Mean (average) HDD	3129
Energy for space heating (ktoe)	3978
Adjusted space heating (ktoe)	$3978 \times (2773/3129)$ = 3525